福建南方路面机械有限公司

Fujian South Highway Machinery Co.Ltd

NFJ1175 Tracked Jaw Crusher

Control system

June 15, 2022

Electrical Control System Manual

NFJ1175

This is the commissioning and operation manual of the electrical system of the NFLG NFJ1175 tracked jaw crusher.

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1 User notice

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This manual contains information necessary to use this equipment. Please read this manual carefully and fully understand it. Before you use this equipment for production activities, our company believes that you have read this manual carefully and can operate and maintain it in accordance with the requirements of this manual. During the use of the equipment, the company cannot guarantee to avoid damage to the equipment caused by man-made sabotage or natural disasters.

The parameters in this manual are only used to describe the production equipment of NFJ1175 tracked jaw crusher. Before publishing, the documents have been carefully checked, but there are unavoidable omissions or inappropriateness. In order to provide better service and technical support, we will sincerely accept any suggestions and comments related to this manual, conduct regular audits, and provide better service and technical support. Make the necessary changes in the next version of the document to meet your needs to the greatest extent possible. We sincerely thank you for your support of our work!

Please treat this manual as a permanent part of the device, always keep it with the device and keep it safe.

2 Precautions for safe use

Before equipment installation, operation, maintenance and overhaul, please be sure to read and understand the relevant chapters in this manual and the mechanical operation manual attached to the equipment, and be familiar with the meanings of all labels, nameplates, safety information and precautions on the equipment before using this equipment, if strictly followed, will be of great help to your security.

2.1 Safety level

In this instruction manual, the severity of safety precautions is divided into three categories: Danger, Warning and Caution to warn you of different potential dangers or remind you to pay attention to relevant information.

Danger:

This represents an imminently hazardous situation and it is assumed that when equipment is incorrectly installed, operated and maintained, it will, if not avoided, result in death or serious injury.

Warning:

This indicates a potentially hazardous situation, which is envisaged when equipment is incorrectly installed, operated and maintained, which, if not avoided, could result in death or serious injury.

Caution:

This indicates that a potentially hazardous situation exists, and it is envisaged that when equipment is incorrectly installed, operated and maintained, it may result in minor to moderate injury, if not avoided, with the possibility of serious consequences depending on the situation.

For the location of all safety signs on the equipment and their meanings, please refer to the mechanical operating manual that came with the equipment.

Note: This equipment is only allowed to be operated and maintained by personnel who have been fully trained by our company's professionals and are qualified to operate. NFLG is not responsible for any consequences arising from failure to follow the provisions of this manual!

2.2 Electrical and engine safety

☆Do not operate the electrical system of the equipment unless you are familiar with the ins and outs of the system and must only be operated by a qualified electrical technician.

☆The electrical system of this equipment must be checked regularly. Damaged cables or loose connectors must be repaired immediately. Only use genuine fuses that meet the specified amperage ratings.

☆Always disconnect the battery ground lead before performing any maintenance work on the electrical system or servicing the engine. When welding work on the equipment, it must be ensured that the battery pack is isolated.

☆Never fill the fuel tank with the engine running, while smoking, or near an open flame.

☆Never short-circuit the positive and negative poles of the battery, as this may cause a fire and damage the electrical system.

☆Batteries contain sulfuric acid, an electrolyte that can cause severe burns and generate explosive gases. Wear personal protective equipment and avoid contact with skin, eyes or clothing during maintenance.

2.3 Equipment operation and maintenance precautions

☆Staff should wear protective equipment necessary for safe production, such as safety helmets, protective gloves, insulating rubber shoes, etc.

☆It is strictly forbidden for untrained and unqualified personnel to start the machine, operate it and take shifts without authorization. Staff with different responsibilities perform their own duties, and it is strictly forbidden to mix and change positions.

☆Before starting the device or before moving the device, check around the device. Make sure that no one is under, on, or near the device. Inform other operators and those present that you are starting or moving equipment and do not activate or move equipment until everyone is evacuated.

☆During the production process of the equipment, all action mechanisms are in operation, and no one is allowed to enter the production site without permission.

☆If there is no need for overhaul and maintenance, ensure that all control cabinet doors are completely closed to prevent dust from entering the cabinet.

☆The equipment site staff should always put safety production in the first place and maintain a sufficient safety distance from various action mechanisms on the equipment.

☆The emergency stop switches at various positions on the equipment should be very familiar, so that when an emergency occurs, the emergency stop operation can be performed as soon as possible.

☆When overhauling, maintaining and maintaining the equipment, the battery isolation switch must be turned off, and the battery isolation switch must be locked at the same time. Before starting work, check that the disconnected parts are not live, grounded, or shorted.

☆Smoking and teasing are strictly prohibited within the scope of the equipment. Non-conductive fire extinguishing equipment (such as dry powder fire extinguisher) necessary for safe fire protection should be equipped around the equipment. Do not use foam to extinguish the fire to avoid electrical short circuit and electric shock accident when using fire extinguishing equipment.

☆According to the actual situation on site, set up equipment safety operation procedures. All personnel involved in the operation and maintenance of the machine must be familiar with the various clauses listed in the safety operation regulations, and strictly abide by them, and resolutely prevent personal injury and equipment accidents caused by misoperation.

☆After the shift work is completed, the staff should remove the ignition switch key, do the final equipment inspection and cleaning, fill in the work log, and cut off the main power switch and battery isolation switch before leaving.

For more safety information, see the mechanical operating manual that came with the equipment.

3 System introduction

3.1 System composition

NFJ1175 tracked jaw crusher control system, the main hardware is composed of high-performance vehicle controller, display screen, node module, internationally renowned low-voltage electrical appliances, complete sets of wiring harnesses and control cabinets and other hardware, combined with the company's self-developed professional control software system to achieve remote control of tracked jaw crusher and automatic production control.

The control system is powered by 24V DC, which is composed of two DC 12V batteries connected in series. The battery group connected in series is charged in real time by the small generator that comes with the engine. The control system takes the vehicle controller as the main station, the display screen and multiple node modules as the slave stations to form the controller area network (CAN for short), the signal acquisition of each sensor, the control of each solenoid valve, etc. are all through the channel of the node module to accomplish. In addition, the high-performance on-board controller also supports the J1939 protocol, which can facilitate communication with the ECU of the engine, and realize the speed control of the engine and the reading of effective information. The matching engine-specific wiring harness and casting-integrated equipment wiring harness provide a stable and reliable line basis for the control system.

3.2 Main feature

☆The main low-voltage electrical appliances adopt world-renowned brands such as ABB, Weidmüller, Rockwell, and Siemens.

☆Adopt IFM vehicle controller imported from Germany: protection level IP67, anti-vibration, external circuit self-diagnosis and other protection functions.

☆The 7-inch IFM color display screen imported from Germany is adopted: the high-resolution, high-performance, all-Chinese control interface brings users an intuitive and responsive operation experience.

☆The complete and comprehensive fault alarm system adopts the form of pictures and texts, and can save the latest 200 fault history records for easy viewing.

☆The system field bus adopts the most advanced CAN communication technology in the world, which is stable and reliable, with strong anti-interference and simple wiring.

☆The equipment wiring adopts heavy-duty connectors\casting integrated connectors to realize signal docking, and the connection is firm and beautiful.

☆Sound and light alarms are given before the equipment is turned on or during walking to remind people around the equipment to pay attention to safety.

☆When the equipment fails, an audible and visual alarm is automatically issued, and there are corresponding protection measures to ensure the safety of the equipment and personal safety.

☆The parameter setting flexibility is high to meet the application requirements of different working conditions, and all parameters can be restored to factory settings with one key.

☆Using the advanced J1939 communication technology, the fault of the EFI engine can be diagnosed on-site.

☆The control of the crushing machine introduces a compensation algorithm, which can ensure that the efficiency of the hydraulic system remains constant in the case of sudden load changes, so that the performance of the main machine is not affected.

4 Electrical installation specifications

(1) The whole machine must have good grounding protection measures, and the grounding resistance is less than or equal to 4Ω.

(2) The wiring should be neatly and reliably fixed to avoid the appearance of coils. The openings reserved for the wire routing in the mechanical part should be covered with broken plastic hoses around the holes to protect the wires and prevent the harness from being worn and scratched which result insulation performance declines. If the damaged wire needs to be replaced, the specification and model of the wire should meet the requirements of the drawing.

(3) Multiple wires should be bundled with cable ties or wrapping tapes, and there should be enough bending length to keep the wire harness free from tension during the movement.

(4) Each connector must be locked/fastened in place. When connecting the connector of the cabinet and the connector of the wiring harness, there should be a one-to-one correspondence. Only after the label is aligned, the connector can be connected.

(5) When wiring, pay special attention to the color of the wire used for the ground wire, and both ends of the wire should have a number.
(6) Before the cabinet is hoisted, the lifting lug bolts should be carefully checked to ensure their firmness. The cabinet must be fixed and locked with a special shock-absorbing pad.
(7) The wiring should make the bend radius as large as possible, avoid sharp corners, and avoid places of friction and high temperature.
(8)All other wiring harness connections must be made before connecting the battery wiring harness to the engine and equipment electrical systems.
(9)The engine wiring harness should be secured to the engine case or engine mounting bracket in a suitable position using cable ties.
⑽ The bolt terminals of the engine should be firmly locked, and must be covered with a suitable pipe end sheath for insulation protection.

5 System debugging

5.1 Preparation before commissioning operation

(1) Before debugging the control system, please read and understand the instruction manual and electrical schematic diagram in detail.
(2) Check whether the wiring of the components in the cabinet is loose, and check whether the connectors of the equipment wiring harness and the engine wiring harness are connected reliably. Before the new machine is powered on or before the engine is started, the wiring terminals of each device and the locked parts of the snap/thread locks of each connector should be Re-tighten. Due to the large vibration of the equipment during operation, the wiring terminals of each device in the cabinet and the connectors of the equipment wiring harness should be regularly reinforced and locked during the daily maintenance process. If the wire is found to be deformed by heat, change color, emit peculiar smell or damage the insulation, it should be replaced immediately. At the same time, it is recommended to clean the dust on the electrical components in the electrical cabinet every month.
(3)Before closing the isolation switch, check the voltage of the engine battery. If the voltage is lower than 24V, the battery should be charged.

5.2 Debugging of main components

(1) Debug instructions:

After the commissioning preparations are ready, the electrical staff and the mechanical commissioning personnel must be present at the same time before the equipment can be powered on and tested. Unless otherwise specified, the debugging of each component is carried out under its own parameter setting interface, and the values are all system default values.

Note: In addition to the debugging of the engine itself, the first advance in the debugging of other components is that the engine must be started.

(2) Control power debugging:

① The reliability of the emergency stop switch must be ensured before power is applied.

② Before debugging the control power supply, check whether the wiring of the DC 24V power supply of each branch is correct according to the drawing, and ensure that the power line number of each branch should be exactly the same as the drawing.

③ After each circuit breaker is closed, check whether the voltage of each power supply output terminal meets the requirements of the actual application standard.

④ After the key switch is powered on, after the warning is over, observe the display screen. If a fault message pops up, please check first, and then perform other operations after the fault is eliminated. For specific fault information and troubleshooting guide, please refer to Chapter 7.

(3)Engine debugging:

① This model uses an EFI engine, and it must be ensured that the communication between the control system and the engine ECU is normal.

② After the key switch is powered on, you should first observe the three indicator lights on the operation cabinet panel (Figure 6.2.3) - from left to right are waiting for running, warning, and shutdown. At this time, the engine ECU will conduct a self-check. The three indicator lights are on at the same time under normal condition, and then go out at the same time after 2-3 seconds. If any indicator light is not off or the three indicator lights are not on at the same time, do not start the engine.

③ The meanings of the three indicators on the control cabinet panel are, from left to right, waiting for operation, warning, and shutdown. If the waiting operation indicator light does not go out, it means that the temperature of the engine intake manifold is too low, and the intake air heater is working. After the heating is completed, the indicator light will automatically go out. If the warning or shutdown indicator light does not go out, it means that the engine is faulty, and an “engine fault” alarm will pop up on the display screen. At this time, you should enter the “engine fault information” interface to check the specific SPN code, FMI code and text information, and then troubleshoot the problem. Except that the spontaneous failure of the engine ECU will limit the start of the engine, other related interlocking conditions include: emergency stop circuit, hydraulic oil level, hydraulic oil temperature and CANBUS bus status. Before the engine is started, it should be ensured that there is no related alarm.

④ Turn the key switch to the ignition gear and hold it. After the engine starts, release the key switch. If the engine fails to start, you should first find out the cause and solve the ignition operation again.

Note: The time of the key switch in the ignition gear should not exceed 10 seconds, and the interval between starting the engine each time should not be less than 5 minutes, otherwise it will lead to the possibility of the starter motor or starter relay being burned out.

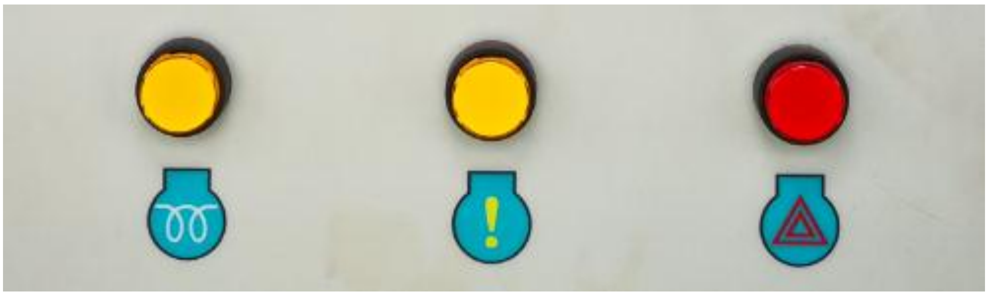


Figure 5.2.3 Engine self-test indicator light

⑤ After the engine is started normally, the state of the engine and hydraulic pressure should be carefully observed. If there is any abnormality, the engine should be turned off immediately and checked. After finding out the cause and solving it, the engine can be restarted.

⑥ After the engine runs normally, let the engine run at idle speed for 3 to 5 minutes, then enter the "System Control Parameters" interface, manually adjust the engine speed, let the speed slowly and steadily increase to the maximum, and run continuously for 20 to 30 minutes.

⑦ If you need to know more about the status information of the engine during the starting or running of the engine, you can enter the "Engine Information" interface to view it at any time or start the engine on this interface.

(4) Auxiliary channel debugging:

① Enter the "System Control Parameters" interface (see Chapter 6.3 for details), and press the [Auxiliary 1 Start and Stop] and [Auxiliary 2 Start and Stop] buttons in sequence. Auxiliary output 1 or auxiliary output 2 status on the display changes from "off" to "on".

② Referring to the drawing, observe the channel output indicator of the corresponding node module, which should be solid yellow.

③ The auxiliary channel is only used as the spare output of the control system. In the standard model configuration, these two channels are not used. If you really need to use it, and it is not used as a separate start-stop interface, you must choose an appropriate start-up method.

(5)Cooling fan debugging:

Enter the "Hydraulic oil temperature control parameters" interface, and press the cooling fan start and stop button. The status of the hydraulic oil cooling fan on the display changes from "off" to "on". Observe whether the hydraulic oil cooling fan operates normally.

Unlike other components, the commissioning of the cooling fan does not require starting the engine first. Because the cooling fan is a high-power component, it is not suitable to run for a long time when the engine is not turned on, generally not more than one minute. If it really needs a long debugging time, please start the engine first, and then carry out the debugging.

In the actual operation of the equipment, once the temperature of the hydraulic oil is greater than or equal to 40° C, after a certain delay, the cooling fan will automatically turn on. On the contrary, once the hydraulic oil temperature is less than 40 ℃, after a certain delay, the cooling fan will automatically turn off. (The temperature value can be tested by the signal simulator output 4-20mA)

(6)Main conveyor belt debugging:

① Before debugging, the tail of the main conveyor belt must be lowered to the lowest level (the folding interface should be entered to perform related operations), otherwise the main conveyor cannot be started and a prompt will appear on the screen. Do not arbitrarily disable the tail proximity sensor function, as doing so may cause unpredictable damage to the main conveyor.

② Manually adjust the engine speed to 1200rpm or above. This is critical to avoid false alarms of "low main conveyor belt speed".

③ At the beginning of commissioning, you must first observe the direction of rotation. If the belt reverses, stop the main conveyor belt immediately, turn off the power supply, and adjust the position of the main conveyor belt hydraulic motor in and out of the oil pipe.

④ Adjust the deviation of the belt accordingly. If there is any abnormality, the main conveyor belt should be stopped immediately.

⑤ After the belt runs normally, observe the speed of the belt conveyor on the display screen, the speed should be ⩾65m/min, if the speed value is too low or there is no speed, the screen will pop up an alarm and stop the main conveyor. Please refer to Chapter 7 for fault details. If the main conveyor belt speed wheel sensor function is disabled, the speed value has nothing to do with the running state of the main conveyor belt.

(7)Debugging of iron remover:

Note: The iron remover is optional and may not be installed on the machine.

① Enter the "Iron Remover Control Parameters" interface, and press the [Iron Remover Start and Stop] button. The status of the iron remover on the display changes from "off" to "on".

② If the iron remover is not installed on the machine, refer to the drawings and observe the channel output indicator lights and solenoid valve connector indicators of the corresponding node modules, which should be solid yellow. If the iron remover is installed on the machine table, you should first observe its direction of rotation. If the iron remover is reversed, stop the iron remover, turn off the power supply, and adjust the position of the inlet and outlet pipes of the iron remover hydraulic motor.

(8)Side belt conveyor debugging:

Note: The side belt conveyor is optional and may not be installed on the machine.

① Enter the "Side Belt Conveyor Control Parameters" interface, and press the [Side Belt Conveyor Start and Stop] button. The side belt conveyor status on the display changes from "off" to "on".

② If the side belt conveyor is not installed on the machine table, refer to the drawings to observe the channel output indicator lights and solenoid valve connector indicators of the corresponding node modules, which should be solid yellow. If the side belt conveyor is installed on the machine table, you should first observe its turning direction. If the side belt conveyor is reversed, stop the side belt conveyor, turn off the power and adjust the position of the hydraulic motor of the side belt conveyor.

(9)Host pull rod debugging:

① If the tie rod pressure is less than the lower limit of the charging pressure (default is 46Bar), the hydraulic system will automatically charge the tie rod with liquid until the tie rod pressure is greater than the upper limit (50Bar by default) and automatically stop charging. The tie rod pressure signal is read in real time, as long as the engine is running, the tie rod filling process will be completed automatically.

Note: If the pressure of the tie rod reaches the upper limit, the moment when the liquid-filled solenoid valve loses power, the pressure of the tie rod rises sharply, indicating that there may be problems in the hydraulic or mechanical parts, and the relevant hydraulic or mechanical parts should be checked.

② In the "folding mode" interface, "automatic mode" interface, "manual mode" interface, "blocking mode" interface or "crushing host control parameter" interface, if the tie rod filling time exceeds 15 seconds, the filling of the tie rod has not been completed, the display screen will pop up an alarm of "filling time is too long" and interrupt the automatic filling process. At this time, an investigation should be carried out to find out the reason why the tie rod cannot be filled with liquid normally.

③ Adjust the jaw gap when the crusher is not running. A pull rod pressure relief button is installed on the side of the hydraulic cabinet. Press and hold the button with one hand and observe that when the pressure on the pull rod pressure gauge drops to close to 0Bar, the other hand operates the corresponding hydraulic handle to adjust the jaw.

Note: If the pressure relief button cannot be normally relieved or the pressure relief is incomplete after operating the pressure relief button according to the above steps, please do not adjust the gap between the jaws. It should be checked first and resolved before proceeding.

⑽Crusher debugging:

①Before commissioning, check whether there is foreign matter in the crushing chamber, check whether the speed proximity switch is installed correctly; ensure that the pressure of the tie rod is normal, and the abnormal pressure of the tie rod will prohibit the start of the crushing machine; adjust the gap of the discharge port of the machine to the specified range.

②Manually adjust the engine speed to 1400rpm or above. This is critical to avoid false alarms of "low speed of the main conveyor belt" or "low speed of the crusher".

③Only after the main conveyor belt is started, the jaw crusher is allowed to start. The operation of the jaw crusher can be either forward or reverse. After the jaw crusher is started, you should first observe the steering of the inertia wheel. Clockwise is forward rotation, and counterclockwise is reverse rotation. The steering should match the direction of the console startup icon pressed on the display. If it is not consistent, stop the jaw crusher, turn off the power and adjust the positions of the two sets of solenoid valve connectors on the swash plate pump (first ensure that the channel wiring on the corresponding node module is correct), and restart the host.

④During the start-up process of the jaw crusher, observe the current values of the swashplate pump 1 and swashplate pump 2 on the display screen. If the current feedback value is not synchronized with the given value, the jaw crusher does not act and the feedback value is less than 200mA, the power should be turned off and check that the wiring is properly and securely connected. Under normal circumstances, the given mA value of the proportional valve is not much different from the feedback mA value.

⑤The debugging speed of the jaw crusher is 50% by default regardless of whether it is forward or reverse. It can be adjusted arbitrarily according to the needs of debugging. It is not recommended to set the speed below 50% for debugging. After the jaw crusher starts normally, if the alarm of "low crusher speed" appears, under the condition of ensuring the correct installation of the proximity switch, it may be caused by the too high number of action pulses set by the proximity switch. Just lower the number of action pulses of the proximity switch, please contact relevant professionals for specific operations.

⑥After the jaw crusher starts normally, no matter whether it is forward or reverse, it needs to be idling for 20-30 minutes continuously to observe whether the operation is stable, whether the vibration is within the normal range, and whether there is any abnormal sound.

⑾ Vibration feeder debugging:

①Unfold the upper part of the vibrating feeder and fix it in place to ensure that there is nothing on the vibrating feeder.

②At the beginning of debugging, the speed value of the feeding belt conveyor should be adjusted to 50%. After starting, if the vibration of the vibrating feeder is abnormal, it should be stopped immediately, the main power supply should be disconnected, and an investigation should be done.

③During the start-up process of the vibrating feeder, observe the current value of the feeder speed on the display screen. If the current feedback value is not synchronized with the given value, the feeder does not move and the feedback value is less than 200mA, the power should be turned off and check that the wiring is properly and securely connected. Under normal circumstances, the given mA value of the proportional valve is not much different from the feedback mA value.

④After the feeder runs normally, adjust the speed of the feeder, the upper limit is 100%, and the lower limit is 10%, observe whether the speed change of the feeder is normal, and measure the amplitude of the feeder with relevant instruments if necessary. Speed 100% corresponds to the parameter value of "maximum speed set point of feeder" under the interface of "Feeder Control Parameters", speed 0% corresponds to the parameter value of "minimum speed set point of feeder belt" under the interface of "Feeder Control Parameters". See chapter 6.3 for details. These two parameter values can be preset according to the actual application.

⑾Tracked walking debugging:

Note: Be sure to raise the tail of the main conveyor belt before the tracked travel operation!

①Enter the "Tracked Walking Mode" interface and select the control mode (wired or wireless). For the specific remote control operation instructions, please refer to chapter 6.6. It is stipulated that the vibrating feeder is the tail and the main conveyor belt is the head. The operator's standing direction should look from the tail to the head. The left hand side is the left forward or backward direction of the tracked, and the right hand side is the right forward or backward direction of the tracked.

②Carry out the track steering test: operate the direction keys on the remote control one by one, and observe whether the movements of the left and right tracks are consistent with the direction keys on the remote control.

③Carry out the crawler walking test: press the left and right forward or left and right backward direction keys at the same time, and observe whether there is any deviation in the process of tracked forward or backward. If a slight deviation is found, it can be corrected by adjusting the given value of the tracked proportional valve. The maximum difference between the left and right direction adjustment should not exceed 50mA. If a serious deviation is found, first check whether the hydraulic or mechanical aspects exist question.

④The wireless remote control device of this model is standard. After the wireless remote control function is fully activated, the green alarm light should be on. At this time, other buttons of the wireless remote control can be used (except the activation button on the remote control).

⑤When the tracked is running, there are two modes for the control of the engine speed——manual and automatic, which can be changed under the "Control System Parameters" interface. The default is to manually control the engine speed. In the ″Tracked Walking Mode″ interface, the speed of the engine can be increased or decreased through the high and low speed buttons, thereby controlling the speed of the tracked steering or walking. If the automatic engine control mode is selected, the engine speed will automatically rise to the speed value set by the ″Tracked High-speed Travel Engine Speed″ parameter after the running warning for 7s ends. Due to the limitations of the workshop site, it is recommended to use the manual mode to change the engine speed for track steering or walking tests to ensure safety.

6 Display operating instructions

6.1 Main menu interface

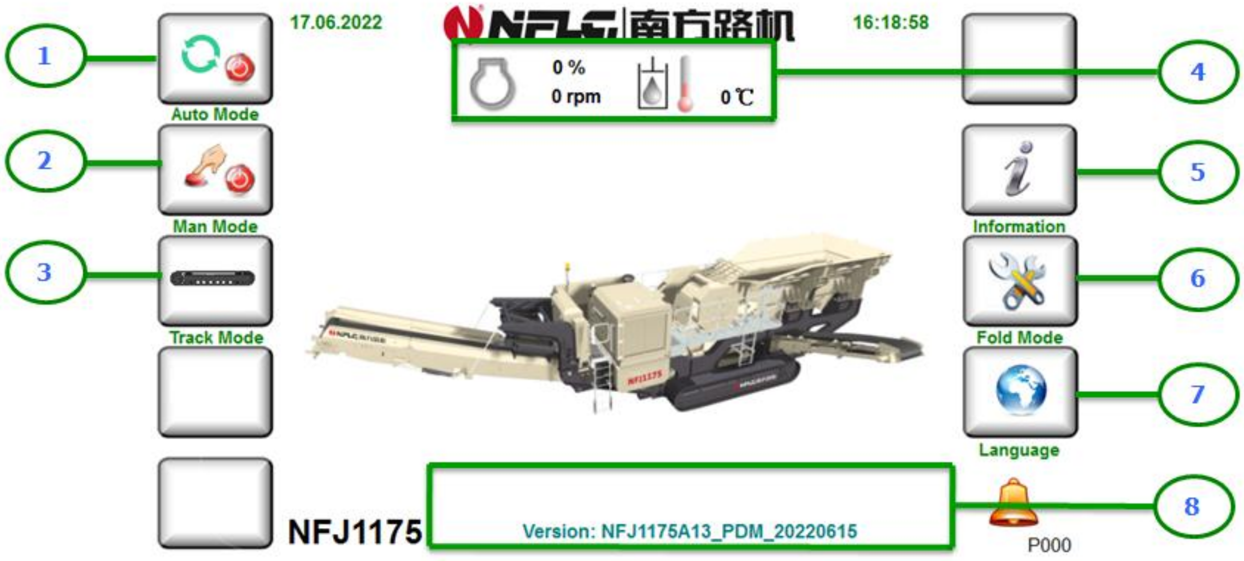

1
2
3
4
5
6
7
8
17.06.2022
NFLG|南方路机
16:18:58
0 %
0 rpm
0℃
Auto Mode
Man Mode
Track Mode
Information
Fold Mode
Language
NFJ1175
Version: NFJ1175A13_PDM_20220615
P000

Figure 6.1.1 "Main Menu" interface

1—Enter the "Automatic Production Mode" interface (see Chapter 6.6 for details)

2—Enter the "Manual Production Mode" interface (see Chapter 6.7 for details)

3—Enter the "Tracked Walking Mode" interface (see Chapter 6.5 for details)

4—Display of engine load percentage, engine speed and hydraulic oil temperature.

5—Enter the "Function Information" interface (see Chapter 6.2 for details)

6—Enter the "Folding Mode" interface (see Chapter 6.9 for details)

7—Enter the "Language Selection" interface (see Chapter 6.10 for details)

8—PDM program version number and specific machine number.

The "Main Menu" interface appears automatically after the system is powered on. During the start-up warning process, all buttons on the screen are not allowed to operate. If the equipment has no fault alarm, after the warning is over, the user can enter the corresponding interface to perform the next operation as needed. If the equipment has a fault alarm, after the warning is over, the interface will automatically jump to the "Alarm Information" interface, listing the current fault alarm information.

6.2 Function information interface

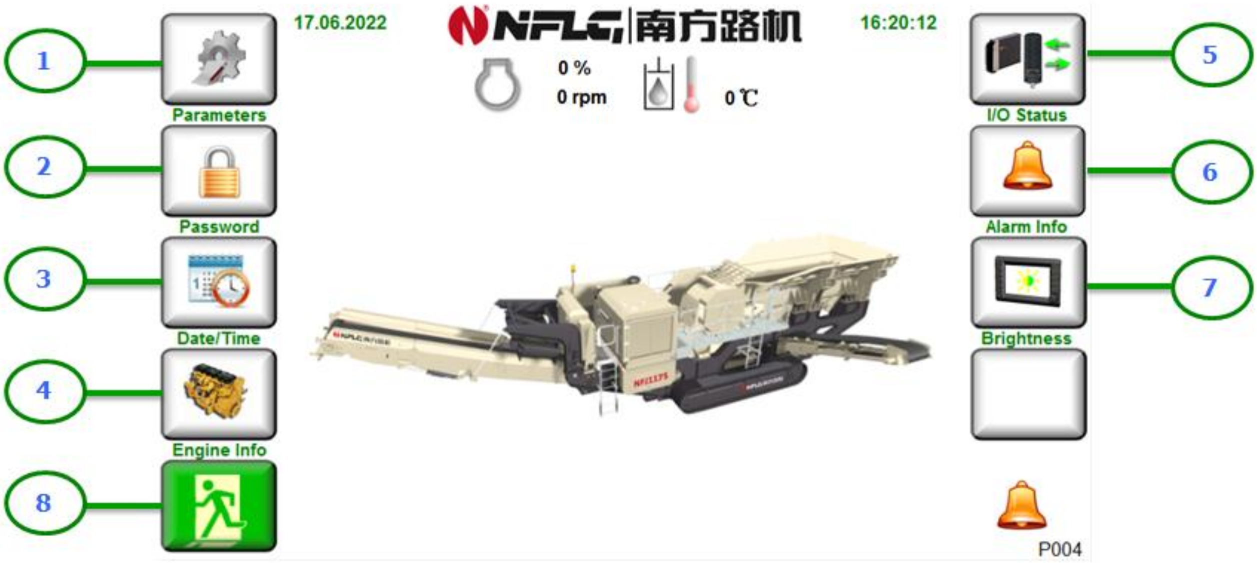

1
2
3
4
8
Parameters
Password
Date/Time
Engine Info
17.06.2022
NFLG 南方路机
16:20:12
0 %
0 rpm
0 ℃
5
6
7
I/O Status
Alarm Info
Brightness
P004

Figure 6.2.1 "Function Information" interface

1—Enter the "Parameter Setting" interface 2—Enter the "Password Input" interface

3—Enter the "Date Settings" interface 4—Enter the "Engine Information" interface

5—Enter the "Node Information" interface 6—Enter the "Alarm Information" interface

7—Enter the "Screen Brightness Adjustment" interface 8—[Exit] button

This interface provides the entry of various important function information of the system.

6.3 Parameter setting interface

In the "Main Menu" interface, select the [Function Information] button (Figure 6.1.1-5), and it will enter the "Function Information" interface, and then select the [Parameter Setting] button to enter the "Parameter Setting" interface, as shown in Figure 6.3.1 shown.

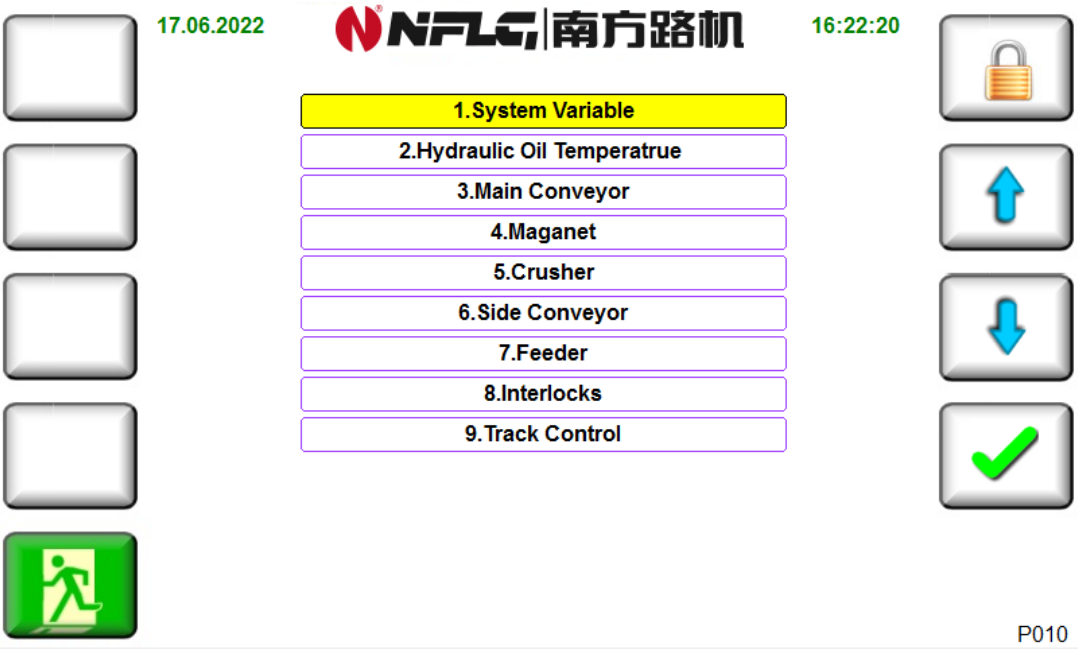
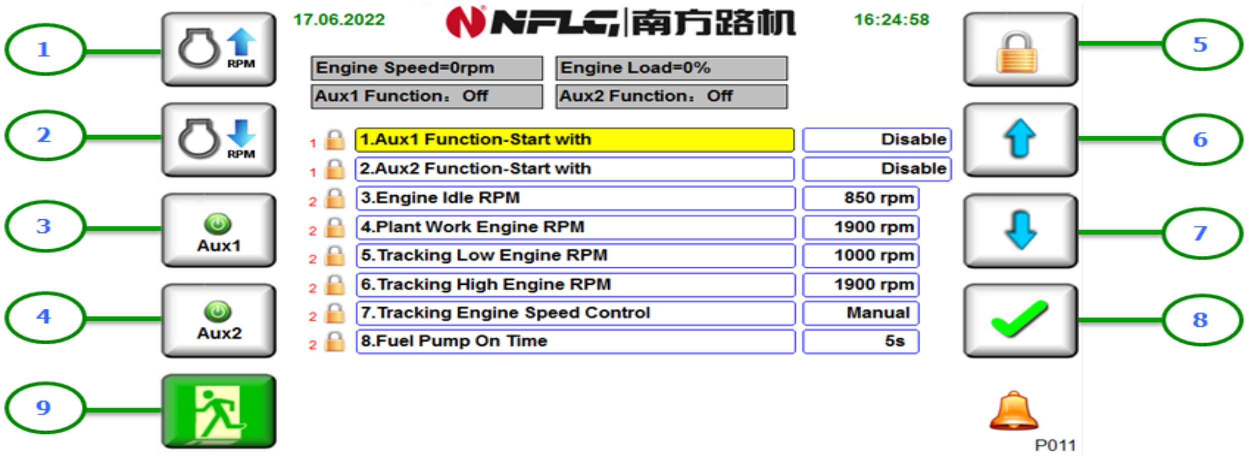

17.06.2022
NFLG 南方路机
16:22:20
1.System Variable
2.Hydraulic Oil Temperatrue
3.Main Conveyor
4.Maganet
5.Crusher
6.Side Conveyor
7.Feeder
8.Interlocks
9.Track Control
P010

Figure 6.3.1 "Parameter Setting" interface

This interface integrates the setting functions of all parameters of the system, and users can make reasonable changes according to different working conditions.

Note: The following content only introduces some parameter interfaces for common operations. Before modifying any parameter, please read the manual carefully, understand and master it thoroughly before proceeding.

(1) System control parameter interface


1
2
3
4
9
RPM
RPM
Aux1
Aux2
17.06.2022
NFLG 南方路机
16:24:58
5
6
7
8
P011

Engine Speed=0rpm
Engine Load=0%
Aux1 Function: Off
Aux2 Function: Off

1	🔒	1.Aux1 Function-Start with	Disable
1	🔒	2.Aux2 Function-Start with	Disable
2	🔒	3.Engine Idle RPM	850 rpm
2	🔒	4.Plant Work Engine RPM	1900 rpm
2	🔒	5.Tracking Low Engine RPM	1000 rpm
2	🔒	6.Tracking High Engine RPM	1900 rpm
2	🔒	7.Tracking Engine Speed Control	Manual
2	🔒	8.Fuel Pump On Time	5s

1—[Engine Speed Increase] button, press and hold to increase the engine speed, the function of this button will not be described below.

2—[Engine Speed Reduction] button, press and hold to reduce the engine speed, the function of this button will not be described below.

3—[Auxiliary 1 Start/Stop] button can start or stop the output of auxiliary channel 1.

4—[Auxiliary 2 Start/Stop] button can start or stop the output of auxiliary channel 2.

5—Enter the "Password Input" interface. If the password is valid, the icon will display the unlocked state. If the password is invalid, the icon will display the locked state. The function of this button will not be described later.

6-The function of the [Parameter Item up Selection] button is equivalent to rotating the knob counterclockwise. The function of this button will not be described below.

7-The function of the [Parameter Item down Selection] button is equivalent to rotating the knob clockwise. The function of this button will not be described below.

8-[Confirm Selection] button, the function is equivalent to pressing the center of the knob, the function of this button will not be described below.

Main parameter setting description:

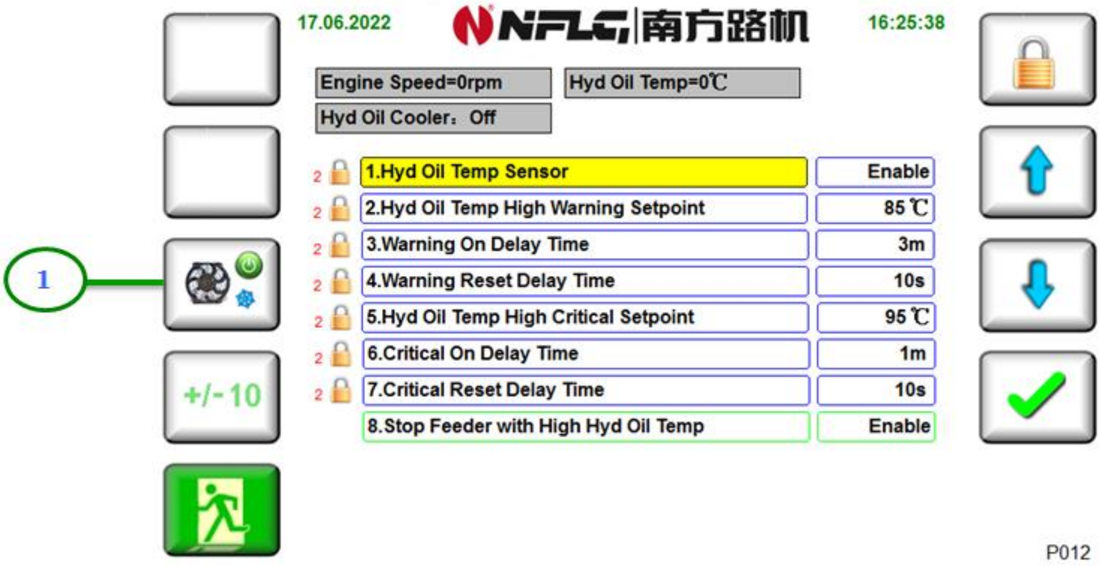
①Start mode selection - auxiliary output 1 - this function parameter is to select the start mode of auxiliary channel 1. You can not use it, or you can choose to follow other components to start at the same time. The startup mode selection of auxiliary output 2 is also the same, and will not be introduced here.

②Engine idle speed, working mode engine speed, tracked low-speed running engine speed, crawler high-speed running engine speed can be reasonably set according to the actual working conditions, and the default values can be used in general. The engine idle speed setting should not exceed 900rpm.

③Tracked travel engine control mode - there are two modes of manual and automatic. When the equipment is running on the "Tracked Walking Mode" interface, if manual is selected, the engine speed can only be adjusted through the [Engine Speed Increase/Decrease] button; If Auto is selected, the engine speed will be automatically adjusted. The engine speed corresponding to the left/right forward and left/right reverse of the unilateral track is the fifth parameter, and the engine speed corresponding to the simultaneous forward and reverse of the bilateral track is the sixth parameter.

④The duration of the engine fuel pump action - the default is 5s. The working time depends on the location of the diesel tank and the length of the oil pipe.

(2) Hydraulic oil temperature control parameter interface



17.06.2022 16:25:38

Engine Speed=0rpm
Hyd Oil Temp=0℃
Hyd Oil Cooler：Off

2	1.Hyd Oil Temp Sensor	Enable
2	2.Hyd Oil Temp High Warning Setpoint	85℃
2	3.Warning On Delay Time	3m
2	4.Warning Reset Delay Time	10s
2	5.Hyd Oil Temp High Critical Setpoint	95℃
2	6.Critical On Delay Time	1m
2	7.Critical Reset Delay Time	10s
	8.Stop Feeder with High Hyd Oil Temp	Enable

1—[Cooling Fan Start and Stop] button can start or stop the output of the cooling fan, which is only valid in this interface.

Main parameter setting description:

①Hydraulic oil temperature sensor function selection - release or maintain the interlocking relationship between hydraulic oil temperature and hydraulic oil temperature sensor status to the control system by disabling or activating, the default value is "active".

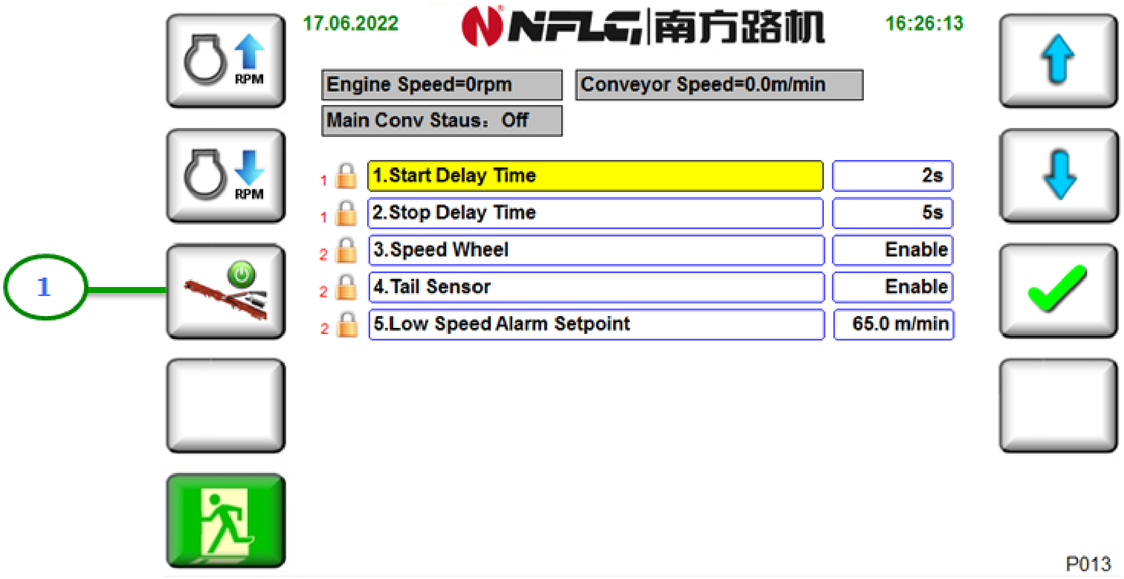
Note: Changing the value of this parameter to "Disabled" may have serious consequences, please use it after careful consideration. This disable function is only applicable when the temperature sensor fails. In order to ensure the normal operation of the equipment and emergency use, please replace the temperature sensor of the same type as soon as possible, and the parameter value must be changed to "active" again.

②Hydraulic oil high temperature warning value - If the hydraulic oil temperature is higher than or equal to this set value for more than the delay time of the third item, the system will pop up an alarm to prompt the operator. This function is valid when the "hydraulic oil temperature sensor function selection" parameter is set to "active".

③Hydraulic oil high temperature shutdown value - if the hydraulic oil temperature is higher than or equal to this set value for longer than the delay time of the sixth item, the system will pop up an alarm and stop the production process immediately. This function is valid when the "hydraulic oil temperature sensor function selection" parameter is set to "active".

⑤Hydraulic oil high temperature stop feeder function selection - disable or activate to release or maintain the interlock relationship to the vibrating feeder when the hydraulic oil temperature exceeds the value set by "Hydraulic oil high temperature warning value", the default value is "active" ".

(3) Main conveyor belt control parameter interface



17.06.2022

16:26:13

Engine Speed=0rpm

Conveyor Speed=0.0m/min

Main Conv Staus：Off

1	1.Start Delay Time	2s
1	2.Stop Delay Time	5s
2	3.Speed Wheel	Enable
2	4.Tail Sensor	Enable
2	5.Low Speed Alarm Setpoint	65.0 m/min

P013

1—[Main conveyor belt start and stop] button can start or stop the output of the main conveyor belt, which is only valid in this interface.

Parameter setting description:

①Start delay time - the waiting time after the main conveyor belt receives the start signal until the solenoid valve is energized. This delay time applies in any operating mode.

②Stop delay time - this delay time is applicable to the sequential stop process under the interface of "automatic production mode" or "manual production mode". After the crusher stops, the main conveyor belt stops the delay countdown.

Under normal circumstances, the default value of the start delay time and stop delay time of the component can be used. When it is really necessary to adjust the parameters to better meet the production requirements, the user can make appropriate changes.

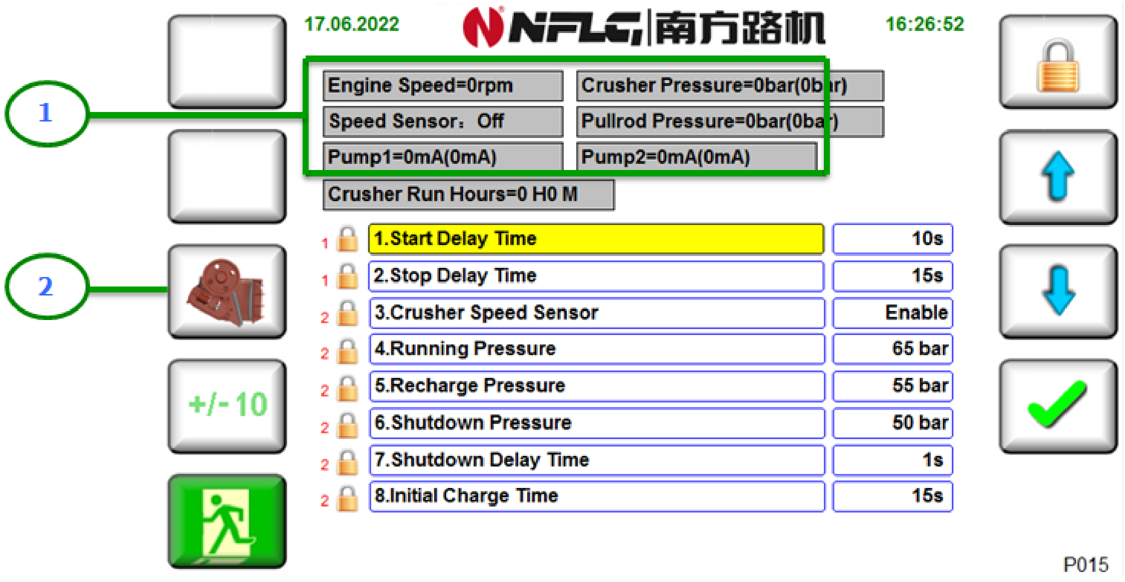
③Speed wheel sensor function selection - release or maintain the interlocking relationship of the speed wheel sensor input signal to the control system by disabling or activating, the default value is "active".

④Tail proximity sensor function selection - disable or activate to release or maintain the interlocking relationship between the input signal of the rear proximity sensor of the main conveyor belt on the tracked running or the main conveyor belt running, the default value is "active".

Note: If the parameter value of the above ③④ is changed to "disabled", it may lead to serious consequences, please use it after careful consideration. The disable function is only applicable to the case of sensor failure. In order to ensure the normal operation of the device and emergency use, please replace the sensor of the same type as soon as possible, and the parameter value must be changed to "active" again.

⑤Main conveyor belt speed low alarm value - if the "speed wheel sensor function selection" is "active", this parameter is valid. When the main conveyor belt is started, if the actual speed is lower than the set value, the system will pop up an alarm and stop the main conveyor.

(4) Crusher control parameter interface



17.06.2022 16:26:52

Engine Speed=0rpm
Crusher Pressure=0bar(0bar)
Speed Sensor：Off
Pullrod Pressure=0bar(0bar)
Pump1=0mA(0mA)
Pump2=0mA(0mA)
Crusher Run Hours=0 H0 M

1	1.Start Delay Time	10s
1	2.Stop Delay Time	15s
2	3.Crusher Speed Sensor	Enable
2	4.Running Pressure	65 bar
2	5.Recharge Pressure	55 bar
2	6.Shutdown Pressure	50 bar
2	7.Shutdown Delay Time	1s
2	8.Initial Charge Time	15s

1—Information bar. It should be noted here that, if the host pressure = 0bar (0Bar), the value displayed in the brackets is the given value of the program, and the value displayed outside the brackets is the feedback value of the sensor. Similarly, the tie rod pressure, swash plate pump 1 and swash plate pump 2 numerical display also follow the same pattern.

2—Enter the interface of "More Parameters of the Host", see Figure 6.3.4.

Parameter setting description:

① Start delay time - the waiting time after the crusher receives the start signal to start the given current output signal. This delay time applies in any operating mode.

②Stop delay time - this delay time is applicable to the sequential stop process in the "automatic production mode" or "manual production mode" interface. After the feeder stops, the crusher will start to stop the delay countdown.

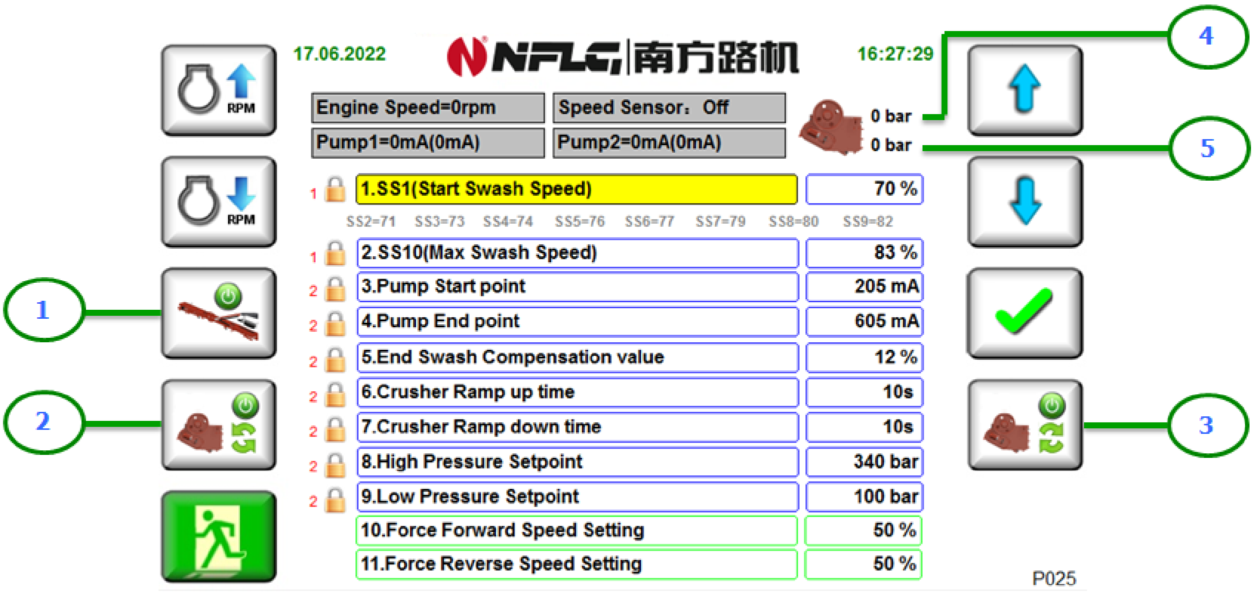
③Crusher speed sensor function selection - by disabling or activating to release or maintain the chain relationship between the speed proximity switch input signal and the crusher, the default value is "active".

Note: Changing the value of this parameter to "Disabled" may have serious consequences, please use it after careful consideration. This disable function is only applicable when the speed proximity switch fails. In order to ensure the normal operation of the equipment and emergency use, please replace the speed proximity switch of the same model as soon as possible, and the parameter value must be changed to "active" again.

④The upper limit of the tie rod filling pressure, the lower limit of the tie rod filling pressure and the pressure value of the tie rod for stopping the crusher are the key parameters for the system to control the tie rod of the crusher. The default values can be used, and it is not recommended for users to change them without authorization.

⑤Delay time for stopping the crushing machine - if the system detects that the pressure of the tie rod is lower than the value set in the sixth item, it will stop the operation of the crushing machine after delaying the time set by this parameter value.

⑥Initial fill duration - The maximum time allowed for the automatic filling process of the drawbar. If the charging of the rod is not completed beyond this time, the system will pop up an alarm and stop the automatic charging.



17.06.2022 NFLG 南方路机 16:27:29

Engine Speed=0rpm
Speed Sensor：Off
Pump1=0mA(0mA)
Pump2=0mA(0mA)
0 bar
0 bar

1	1.SS1(Start Swash Speed)	70 %
	SS2=71 SS3=73 SS4=74 SS5=76 SS6=77 SS7=79 SS8=80 SS9=82	
1	2.SS10(Max Swash Speed)	83 %
2	3.Pump Start point	205 mA
2	4.Pump End point	605 mA
2	5.End Swash Compensation value	12 %
2	6.Crusher Ramp up time	10s
2	7.Crusher Ramp down time	10s
2	8.High Pressure Setpoint	340 bar
2	9.Low Pressure Setpoint	100 bar
	10.Force Forward Speed Setting	50 %
	11.Force Reverse Speed Setting	50 %

Figure 6.3.4 "crusher more parameters" interface

1—[Main Conveyor Belt Start and Stop] button can start or stop the output of the main conveyor belt, which is only valid in this interface.

2—[Start and Stop of Crusher Reverse Rotation] button can start or stop the crusher to make it run in reverse, which is only valid in this interface.

3—[Start and Stop of Forward Rotation of Crusher] button can start or stop the crusher to make it run forward, which is only valid in this interface.

Parameter setting description:

Note: The parameters of this parameter interface are not allowed to be changed by the user without authorization!

① SS1 and SS10 correspond to the minimum and maximum speeds among the 10 adjustable speeds of the crusher in the "Automatic Production Mode" and "Manual Production Mode" interface, and the other speeds SS2 to SS9 are automatically calculated by the system.

②The initial value of the swash plate pump - corresponding to 0% of the swash plate speed - when the crusher starts, the given mA value of the swash plate opening starts from the current value, not from 0 mA. This parameter value is based on the electrical characteristics of the proportional valve.

③The initial value of the swashplate pump - corresponding to 100% swashplate speed - the maximum given mA value to limit the swashplate opening.

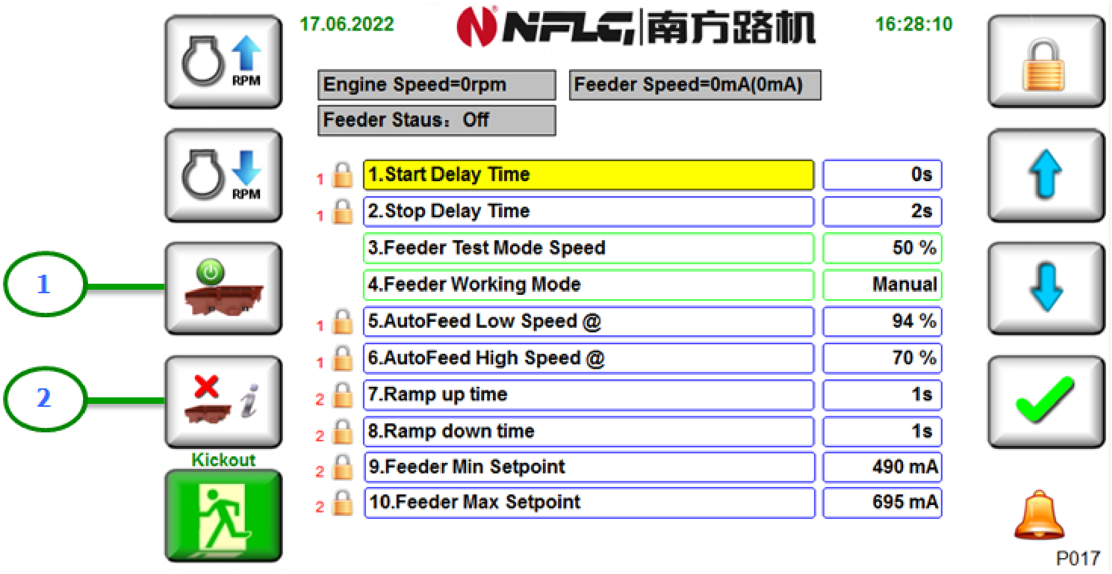
④The combination of the terminal compensation value of the swash plate pump, the high setting value of the swash plate opening compensation pressure and the low setting value of the compensation pressure of the swash plate pump can realize the compensation calculation of the efficiency loss of the swash plate pump. The crusher keeps running at a constant speed at all times during the production process to avoid the phenomenon of falling speed.

⑤Forced crusher forward and reverse speed setting values - these two parameter values are only used for the debugging of the crusher. They are valid under this interface. During the debugging of the crusher, reasonable adjustments can be made according to the needs.

⑥Ramp-up time of swash plate pump—when the crusher starts, the time required for the proportional valve that controls the opening of the swash plate pump to rise from 0mA to a given mA value, so as to realize the soft start of the crusher.

⑦Ramp-down time of swash plate pump—when the crusher starts, the time required for the proportional valve that controls the opening of the swash plate pump to drop from a given mA value to 0mA, so as to realize the soft stop of the crusher.

(5) Feeder control parameter interface



17.06.2022 NFLG南方路机 16:28:10

Engine Speed=0rpm

Feeder Speed=0mA(0mA)

Feeder Staus： Off

1	1.Start Delay Time	0s
1	2.Stop Delay Time	2s
	3.Feeder Test Mode Speed	50 %
	4.Feeder Working Mode	Manual
1	5.AutoFeed Low Speed @	94 %
1	6.AutoFeed High Speed @	70 %
2	7.Ramp up time	1s
2	8.Ramp down time	1s
2	9.Feeder Min Setpoint	490 mA
2	10.Feeder Max Setpoint	695 mA

1—[Start and stop of feeder] button can start or stop the output of the feeder, which is only valid in this interface.

2—Enter the interface of "Pause Parameters of Feeder", see Figure 6.3.5.

Parameter setting description:

①Start delay time - the waiting time for the feeder to start the given current output signal after receiving the start signal. This delay time applies in any operating mode.

②Stop delay time - this delay time is applicable to the sequential stop process in the "automatic production mode" or "manual production mode" interface. After the automatic stop button is pressed, the feeder stop delay countdown starts.

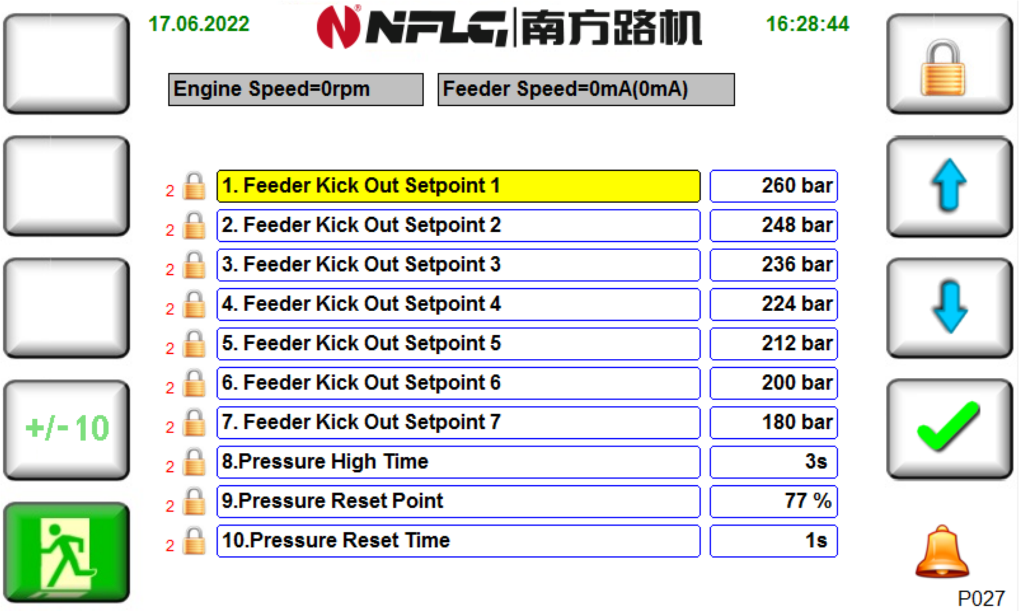
③Feeder test mode speed - This parameter value is only used for the debugging of the feeder, and it is valid under this interface. During the process of debugging the feeder, it can be adjusted reasonably according to the needs.

④Feeder working mode selection - choose whether the speed of the feeder is manually given or automatically given, if it is automatically given, the speed of the feeder will be automatically adjusted within a certain speed range according to the pressure of the crusher. The general principle is as follows:

When the pressure percentage of the crusher is greater than the parameter value set in the fifth item, the feeder speed will run at the mA value set in the ninth item. On the contrary, when the crusher pressure percentage is less than the parameter value set in the sixth item , the feeder speed will run at the mA value set in item 10. Then, when the crushing host pressure percentage is between the fifth parameter and the sixth parameter, the change of the feeder speed is inversely proportional to the change of the crushing host pressure percentage and changes linearly. In general, the default values of these parameters can be used.

⑤Feeder Speed Ramp Time - The time it takes for the proportional valve to ramp from 0 mA to a given mA value when the feeder starts.

⑥ Feeder Speed Ramp Down Time - The time it takes for the proportional valve to ramp down from a given mA value to 0 mA when the feeder starts.



17.06.2022 16:28:44

Engine Speed=0rpm

Feeder Speed=0mA(0mA)

1. Feeder Kick Out Setpoint 1	260 bar
2. Feeder Kick Out Setpoint 2	248 bar
3. Feeder Kick Out Setpoint 3	236 bar
4. Feeder Kick Out Setpoint 4	224 bar
5. Feeder Kick Out Setpoint 5	212 bar
6. Feeder Kick Out Setpoint 6	200 bar
7. Feeder Kick Out Setpoint 7	180 bar
8.Pressure High Time	3s
9.Pressure Reset Point	77 %
10.Pressure Reset Time	1s

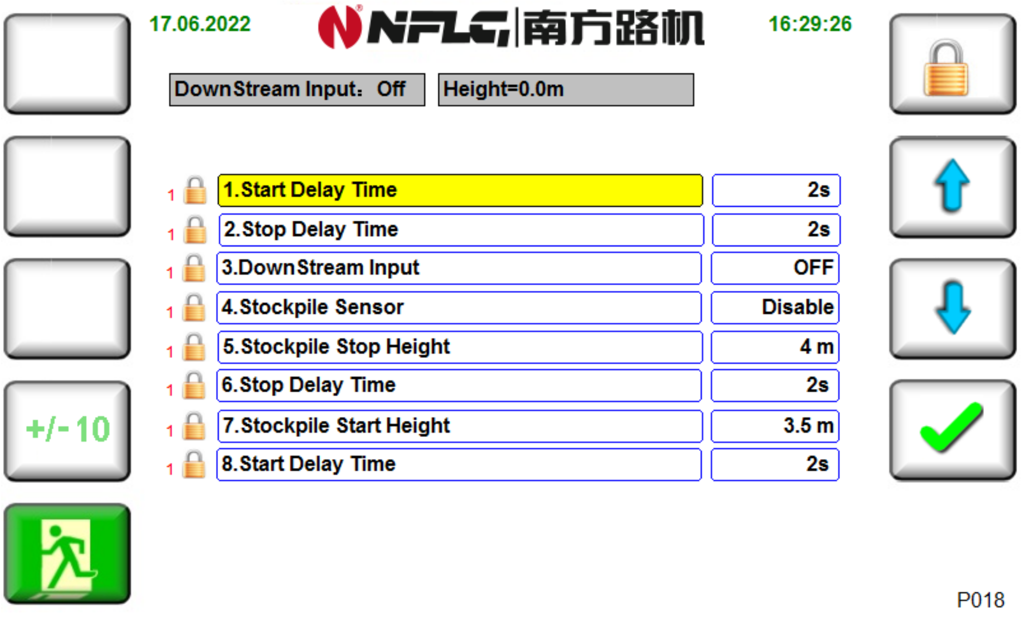
Figure 6.3.5 "Feeder Pause Parameters" interface

Parameter setting description:

The parameters of this interface are the functions of automatic pause and automatic restart of the feeder. According to the characteristics of different stone materials, the system has seven applications to choose from. Each application corresponds to different crusher pressure setting values. Once the crusher pressure exceeds the set value and lasts for a certain time (the eighth parameter), the feeder will automatically pause, the feeder will restart automatically only when the pressure of the crusher returns to the allowable range of the pressure set value of the crusher (parameter 9) and lasts for a certain period of time (parameter 10).

The pause function of the feeder will work regardless of whether the feeder's working mode is manual or automatic.

(6) Upper and lower chain control parameter interface



17.06.2022 NFLG 南方路机 16:29:26

DownStream Input: Off | Height=0.0m

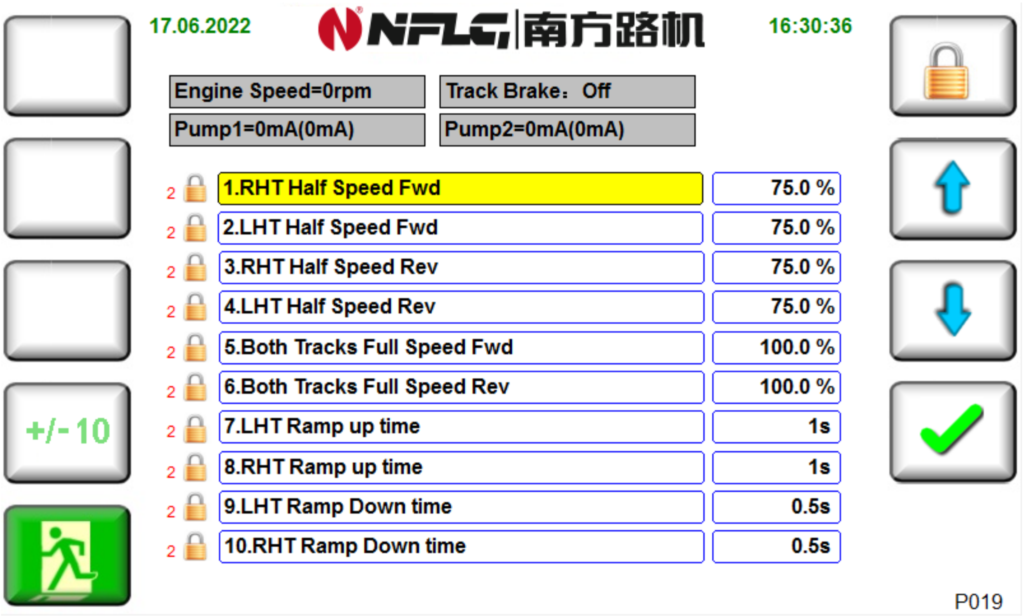
1.Start Delay Time	2s
2.Stop Delay Time	2s
3.DownStream Input	OFF
4.Stockpile Sensor	Disable
5.Stockpile Stop Height	4 m
6.Stop Delay Time	2s
7.Stockpile Start Height	3.5 m
8.Start Delay Time	2s

Parameter setting description:

① Subordinate equipment input function selection - there are "normally open" and "normally closed" options, which can be set according to the signal type connected to the subordinate equipment. It is recommended to activate this function if there is a lower-level device produced in conjunction with this device. In the case of normal external line connection, once the lower-level equipment fails and stops, the system receives the stop signal of the lower-level equipment, and after a certain delay (the second parameter), it will stop the feeder of the equipment, which can prevent continuous supply of material from causing unpredictable impact on the lower equipment.

② Main conveyor belt stack sensor function - This sensor is optional, it is used to measure the height of the material under the main conveyor head, and then control the automatic start or stop of the feeding belt. The parameter settings of the fifth to eighth items are set according to the actual needs of different working conditions.

(7) Tracked walking control parameter interface



17.06.2022　16:30:36

Engine Speed=0rpm　Track Brake：Off

Pump1=0mA(0mA)　Pump2=0mA(0mA)

2	1.RHT Half Speed Fwd	75.0 %
2	2.LHT Half Speed Fwd	75.0 %
2	3.RHT Half Speed Rev	75.0 %
2	4.LHT Half Speed Rev	75.0 %
2	5.Both Tracks Full Speed Fwd	100.0 %
2	6.Both Tracks Full Speed Rev	100.0 %
2	7.LHT Ramp up time	1s
2	8.RHT Ramp up time	1s
2	9.LHT Ramp Down time	0.5s
2	10.RHT Ramp Down time	0.5s

Parameter setting description:

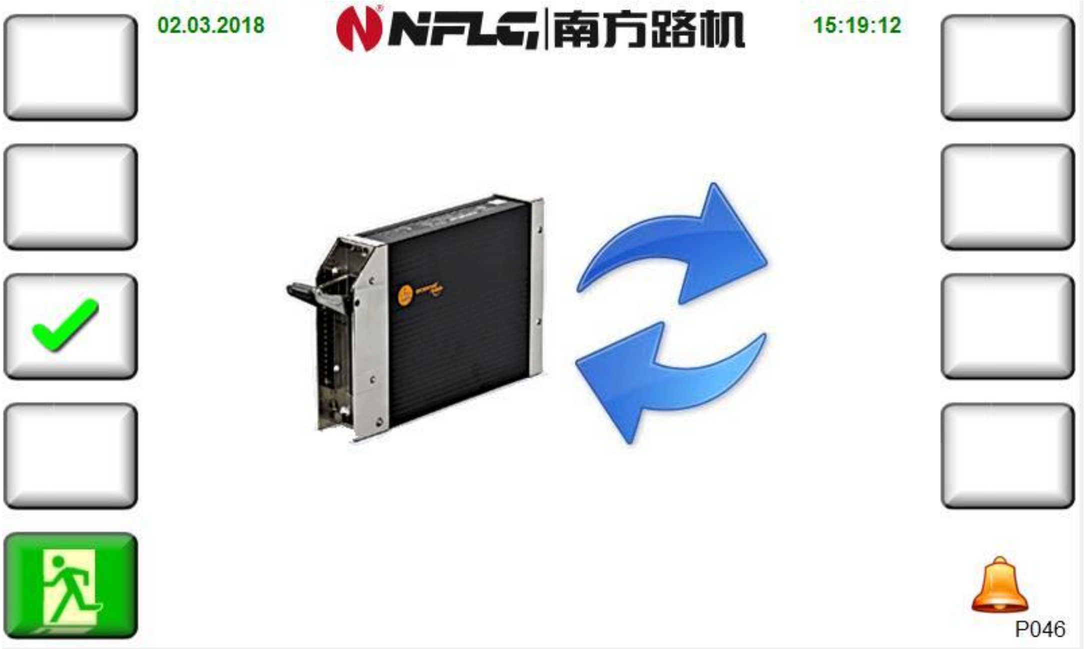
① The first to fourth items are the output of the given current percentage when the left/right single tracked moves forward and backward. The size of the given current percentage directly affects the speed of the crawler walking or turning. The fine-tuning of these values can ensure the consistency of the walking speed of the left and right tracks.

②Left and right tracked tracks forward/backward at high speed - press the button on the remote control for simultaneous forward/backward movement of dual tracks, and the system will output the current of the proportional valve of the swash plate pump with this set value.

③Left/right crawler low-speed walking ramp acceleration time, left and right crawler high-speed walking ramp acceleration time - the time required for the given current of the proportional valve of the swash plate pump to increase from 0mA to the set value, to realize the soft-start function of crawler walking, and at the same time it is also to reduce the impact of instantaneous pressure on proportional valves and related hydraulic systems. This time cannot be set too short, generally the default value can be used.

④Crawler stop ramp down time - the time required for the given current of the proportional valve of the swash plate pump to decrease from the set value to 0mA, to realize the soft stop function of the crawler. Due to the complex and changeable working conditions, from a safety point of view, this setting value is 0.1 seconds. Users cannot make changes without authorization.

⑾ Restore factory default interface



This function is only available when the correct secondary password is entered.

Note: The data for parameter initialization is set by our company based on design principles and experience values, and may not be the best parameter data for the actual operation of the equipment. After the parameter initialization is executed, all the parameters set by the user will be set to the factory default values! Therefore, before executing parameter initialization, please make sure to back up the original data to prevent data loss after initialization. The company is not responsible for equipment damage, software crash, production stagnation and other losses caused by the user's illegal and illegal operations!

6.4 Node information interface

In the "Function Information" interface, select the [I/O Status] button to enter the "Node Information" interface, as shown in Figure 6.4.1.

This interface shows the connection status of all nodes in the entire CANBUS network, and you can choose to enter the node I/O interface you want to view according to your needs. The following points need to be explained:

① "✗" Indicates that the connection between the corresponding node and the PLC is abnormal, "✓" Indicates that the connection between the corresponding node and the PLC is normal.

② In the node I/O status interface, if it is a switch value, the gray light indicates that there is no input or output signal, and the green light indicates that an input or output signal is detected; if it is an analog value, you can view the original value of the signal input.

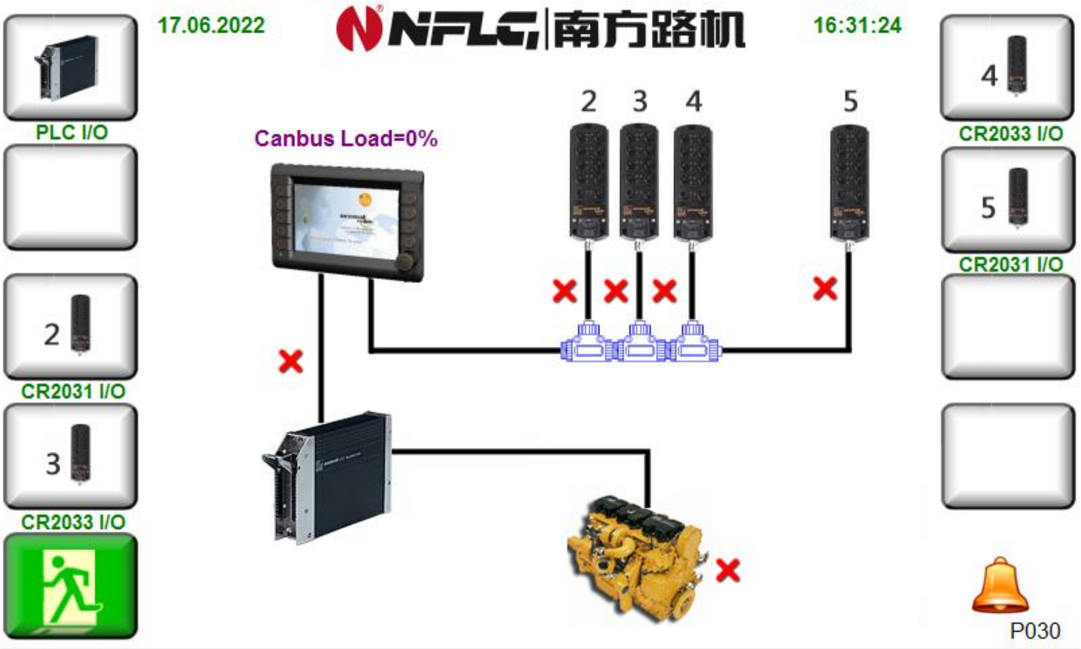
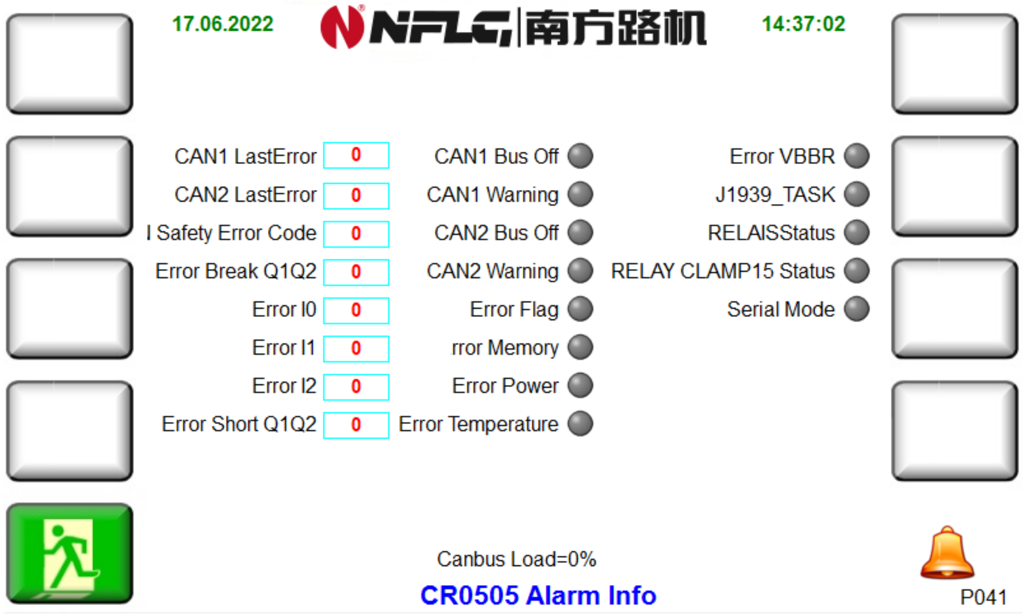

17.06.2022
NFLG 南方路机
16:31:24
PLC I/O
Canbus Load=0%
2
3
4
5
2
CR2031 I/O
3
CR2033 I/O
4
CR2033 I/O
5
CR2031 I/O
P030

Figure 6.4.1 "Node Information" interface

③ In the "Node Information" interface, select the [PLC I/O] button to enter the "PLC I/O Status" interface, and then select the [PLC Diagnosis] button to enter the "PLC Diagnosis" interface, as shown in Figure 6.4.2 As shown, the "PLC Diagnosis" interface covers all the information of PLC fault codes and fault bits. Note: After the "PLC fault!" alarm occurs, the key switch cannot be powered off. It is necessary to enter the "PLC Diagnosis" interface to view the specific PLC fault information and record it, which is convenient for subsequent analysis and troubleshooting.


17.06.2022
NFLG 南方路机
14:37:02

CAN1 LastError 0
CAN2 LastError 0
I Safety Error Code 0
Error Break Q1Q2 0
Error I0 0
Error I1 0
Error I2 0
Error Short Q1Q2 0

CAN1 Bus Off ●
CAN1 Warning ●
CAN2 Bus Off ●
CAN2 Warning ●
Error Flag ●
rror Memory ●
Error Power ●
Error Temperature ●

Error VBBR ●
J1939_TASK ●
RELAISStatus ●
RELAY CLAMP15 Status ●
Serial Mode ●

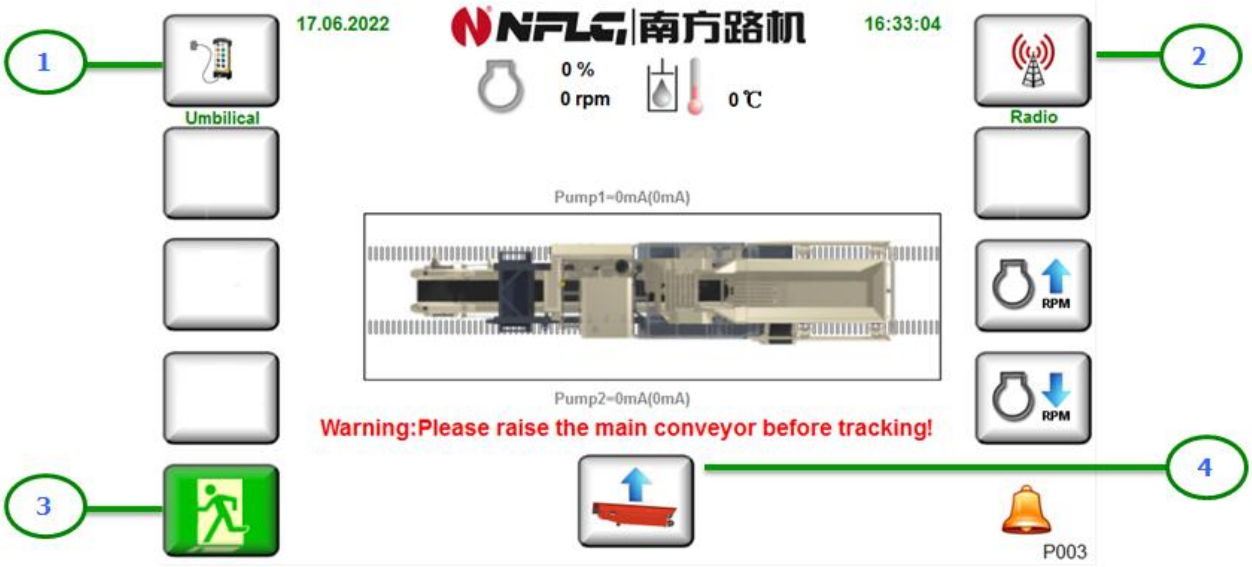
Canbus Load=0%
CR0505 Alarm Info

P041

Figure 6.4.2 "PLC Diagnosis" interface

6.5 Tracked walking mode interface

In the "Main Menu" interface, select the [Walking Mode] button (Fig. 6.1.1-3) to enter the "Crawler Walking Mode" interface, as shown in Fig. 6.5.1. When you want to move the device, you must enter this interface to perform related operations.


1
17.06.2022
NFLG 南方路机
16:33:04
2
Umbilical
0 %
0 rpm
0℃
Radio
Pump1=0mA(0mA)
RPM
RPM
Pump2=0mA(0mA)
3
4
P003

Warning:Please raise the main conveyor before tracking!

Figure 6.5.1 Tracked Walking Mode

1—[Wired Remote] selection button

2—[Wireless Remote Control] selection button

3—[Exit] or [Function Information] button

4—Indicates that the main conveyor belt is not in the correct position

(1)Wired remote control operation

① Press the [Wired Remote] button, the button icon will appear "✓", indicates that the wired remote control function has been selected.

② Make sure that the plug of the wired remote control is correctly connected to the corresponding socket of the cabinet, as shown in Figure 6.5.2-F.

③ Press the button of Figure 6.5.2—E, the wired remote control is activated, and the buzzer and the yellow alarm light simultaneously act to start an early

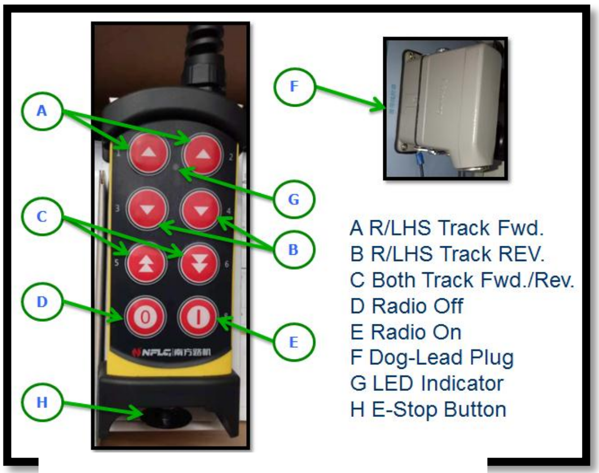

A
F
G
C
B
D
E
H
A R/LHS Track Fwd.
B R/LHS Track REV.
C Both Track Fwd./Rev.
D Radio Off
E Radio On
F Dog-Lead Plug
G LED Indicator
H E-Stop Button

Figure 6.5.2 Wired remote

warning.

④ There needs to be an early warning time of 7 seconds. **During the early warning process, do not operate the buttons on any wired remote control, otherwise the warning will be re-timed.** After the warning is over, the indicator light 6.5.2-G changes from red to green.

⑤ Control the walking or turning of the crawler by pressing the 6.5.2-A, B or C direction keys in the figure.

⑥ After the operation is completed, press the button in Figure 6.5.2-D to turn off the function of each button of the wired remote control, the indicator light 6.5.2-G changes from green to red, and then press the [Wired Remote Control] button again on the display screen , the icon “✓” disappears, indicating that the wired remote control function is disabled.

(2) Wireless remote control operation

Note: The ON\OFF switch of the back shell of the transmitter should be in the ON position before using it, otherwise the transmitter will not start.

① Start the transmitter (Figure 6.5.3) - within the effective range of the wireless signal, make sure to press the stop button, and then pull out the stop button, the top indicator light on the transmitter lights up (green means the battery is fully charged, red means the battery level insufficient), the indicator lights SL and SR flash (red), press the two power-on buttons at the same time, hold for at least 1 second, then the indicators SL and SR light up (red), and finally release the two power-on buttons at the same time, the indicator light SL And the SR goes out, the top indicator light flashes (green), the transmitter is started up.

② Press the [Wireless Remote] button, “✓” will appear on the button icon, indicating that the wireless remote control function has been selected.

③ Press the button of Figure 6.5.3-E, the function of each button of the wireless remote control is activated, and the buzzer and the alarm light will act at the same time to start an early warning. At the same time, the green alarm light on the device illuminates, indicating that the wireless remote control is active and within range.

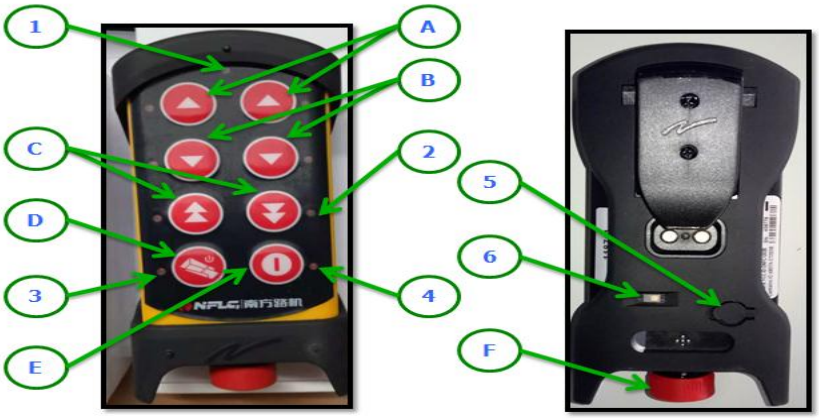

1
A
B
C
2
D
3
4
E
5
6
F
A R/LHS Track Fwd.
B R/LHS Track REV.
C Both Track Fwd./Rev.
D Feeder On/Off
Or Power On Button
E Radio On/Off
Or Power On Button
F E-Stop Button
1 Top LED Indicator
2 Button Indicator
3 Indicator SL
4 Indicator SR
5 Charger Plug
6 On/Off Switch

Figure 6.6.3 Wireless transmitter

④There needs to be a 7-second warning. During the warning process, do not operate any buttons on the remote control, otherwise the warning will be re-timed. ⑤After the warning is completed, the walking or turning of the crawler can be controlled by pressing the direction keys of Figure 6.5.3-A or B.

⑥ After the operation is completed, press the button in Figure 6.5.3-E again to disable the function of each button of the wireless remote control, and then press the [Wireless Remote Control] button again on the display screen. At this time, the icon "✓" disappears, indicating that the wireless remote control function is closed.

Note: After the transmitter has not been operated for a period of time (default 6 minutes), it will automatically turn off the transmitter to save power. Therefore, after the operation of the transmitter is completed, please perform the operation of step ⑥ as soon as possible to avoid the false alarm of "emergency stop failure of the wireless remote control", which will cause the equipment to stop. The protection chain of the emergency stop button to the equipment is only effective when the functions of the buttons of the wireless remote control are activated.

⑦ If you do not operate the wireless remote control for a long time, please toggle Figure 6.6.3-6 to the off position to turn off the power of the remote control.

6.6 Automatic production mode interface

In the "Main Menu" interface, select the [Auto Mode] button to enter the "Auto Production Mode" interface, as shown in Figure 6.6.1.

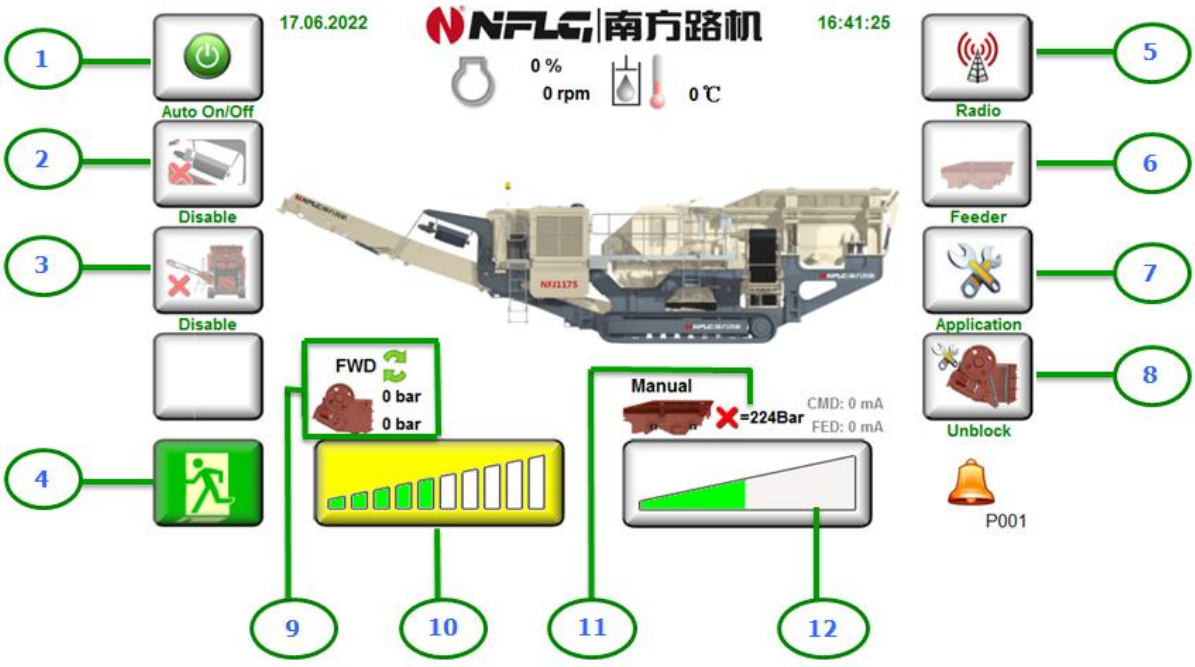

17.06.2022
16:41:25
0 %
0 rpm
0℃
1
Auto On/Off
2
Disable
3
Disable
4
5
Radio
6
Feeder
7
Application
8
Unblock
FWD
0 bar
0 bar
Manual
=224Bar
CMD: 0 mA
FED: 0 mA
P001
9
10
11
12

Figure 6.6.1 Automatic production mode

In this mode, each component starts and stops in a certain order and logical relationship. If the upper-level component is not started, the lower-level component will not start; and the stop of any upper-level component will cause all lower-level components to stop.

The following will describe the icon information, status information, and function buttons in this interface in detail, and the numbers correspond to the labels on Figure 6.6.1:

1—The [Auto start and stop] button is responsible for the start and stop of the automatic production process of the equipment. If there is a related fault alarm, please find out and remove the fault before starting the equipment, otherwise the equipment will not start normally (for details of faults, please refer to Chapter 7).

2—"Iron remover status" icon, if the iron remover function is activated, the icon will be highlighted (Figure 6.6.2-2); if the iron remover function is disabled, the icon will be grayed out and marked with a red cross. After the equipment production process is started, this function button is in an invalid state, and

no operation is allowed, so the activation or deactivation of the iron remover function operation can only be performed before the equipment production process is not started. When activated, the start and stop of the iron remover will be linked to the production process of the equipment; when disabled, the system will ignore the control of the iron remover.

3—"Side belt conveyor status" icon, if the side belt conveyor function is activated, the icon will be highlighted (Figure 6.6.2-3); if the side belt conveyor function is disabled, the icon will be grayed out and marked with a red cross. **After the equipment production process is started, this function button is in an invalid state, and no operation is allowed, so the activation or deactivation of the side belt conveyor function operation can only be performed before the equipment production process is not started.** When activated, the start and stop of the side belt conveyor will be interlocked into the automatic production process of the equipment; when disabled, the system will ignore the control of the opposite side belt conveyor.

4—[Exit] or [Function Information] button, if the automatic production process of the equipment is not started, you can exit this interface and go to the "main menu" interface. When the automatic production process of the equipment starts, this button will automatically change to the [Function Information] button (Figure 6.6.2-1), from which you can enter the "Function Information" page (see Section 6.2).

5—Press the [Wireless Remote] button, "✓" will appear on the button icon, indicating that the wireless remote control function has been selected. Press the button of Figure 6.6.3—E, the function of each button of the wireless transmitter is activated, and the green alarm light on the device lights up, indicating that the wireless remote control is active and within the effective range. In this mode, the wireless transmitter allows remote control of the start and stop of the feeder and the adjustment of the feeder speed. Please refer to chapter 6.5 for the start of the wireless transmitter and the introduction of each button.

6—[Feeder start and stop] button to manually control the start and stop of the feeder. Even in automatic production mode, the feeder must be started manually. The gray display of the button icon indicates that the starting conditions of the feeding belt have not been met (Figure 6.6.1-5), and manual start is only allowed under the premise of fully satisfying the relevant interlock. At this time, the button icon changes to Figure 6.6.2-4 shown.

7—[Application selection] button, you can enter the "application selection" interface, for different stone properties, the system has seven applications to choose from.

8—[Clog Clear] button to enter the interface of "Block Clear Mode", see chapter 6.9 for details.

9—Display the relevant information of the crushing machine - the turning direction of the main machine (can be switched in the "blocking removal mode" interface), the upper Bar value is the crushing machine pressure, and the lower Bar is the main machine tie rod pressure.

10—Crushing host speed adjustment bar, there are ten speeds to choose from. Corresponding to the parameter values of SS1 to SS10 under the interface of "Crushing host control parameters".

11—Automatically suspend the crushing main pressure value of the feeder, and choose different applications (seven options), this value is also different.

12—Feeder speed adjustment. When the working mode of the feeder is "manual", the speed can be adjusted freely manually through the knob. If the working mode is "Auto", only the automatic adjustment range of the feeder speed can be set, 0% corresponds to the ninth item of the "Feeder control parameters" interface, and 100% corresponds to the tenth item of the "Feeder control parameters" interface. Parameter value setting, please refer to chapter 6.3.

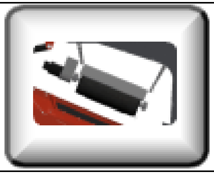
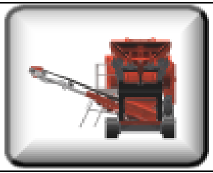
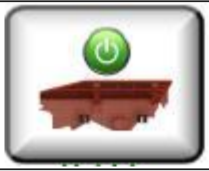
			
1	2	3	4

Figure 6.6.2

6.7 Manual production mode interface

In the "Main Menu" interface, select the [Auto Mode] button to enter the "Manual Production Mode" interface, as shown in Figure 6.7.1.

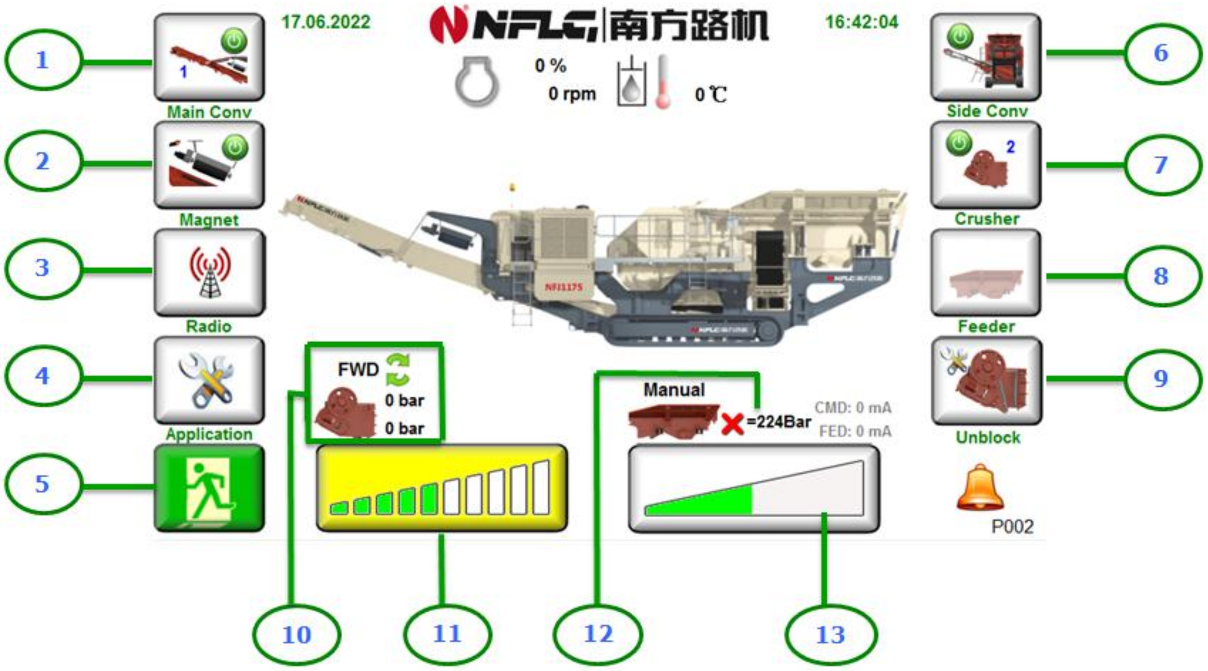

17.06.2022
16:42:04
0 %
0 rpm
0 ℃
1
Main Conv
2
Magnet
3
Radio
4
Application
5
6
Side Conv
7
2
Crusher
8
Feeder
9
Unblock
P002
FWD
0 bar
0 bar
Manual
=224Bar
CMD: 0 mA
FED: 0 mA
10
11
12
13

Figure 6.7.1 Manual production mode

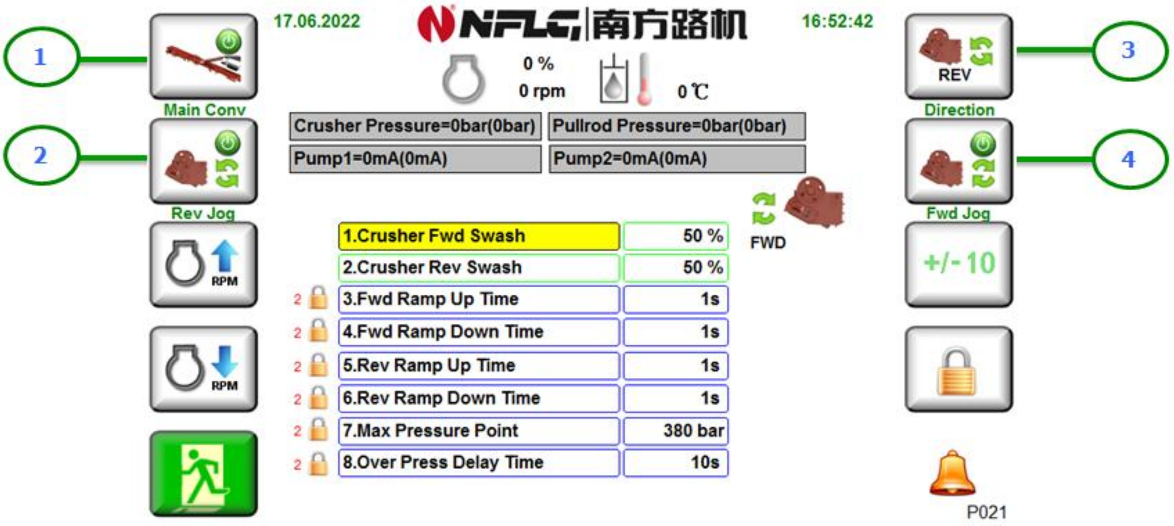
1—[Main conveyor belt start and stop] button
2—[Iron remover start and stop] button
6—[Side belt conveyor start and stop] button
7—[Broken host start and stop] button
8—[Feeder start and stop] button

For the introduction of other icon information, status information, and function buttons, please refer to the content of the "Automatic Production Mode" interface, see chapter 6.6 for details. In this mode, the start and stop of each component need to be manually operated, and there is a certain sequence and logical relationship. If the upper-level component is not started, the lower-level component will not start; and the stop of any upper-level component will cause all lower-level components to stop. The starting sequence of each component is: main conveyor belt -> iron remover (can be ignored when not in use) -> side belt conveyor (can be ignored when not in use) -> crushing main machine -> feeder, and the stop sequence is just opposite to the start sequence.

6.8 Blockage clearing mode interface

In the interface of "automatic production mode" or "manual production mode", select the button of [blocking clear] to enter the interface of "blocking clearing mode", as shown in 6.8.1.

The function of this blockage removal is not a conventional method to clear the blockage in the crusher chamber, but only an auxiliary method. Under certain conditions, through the repeated operation of the forward and reverse jog of the host, the blockage in the crusher chamber can be quickly and effectively removed.



17.06.2022

16:52:42

0 %
0 rpm
0 ℃

Crusher Pressure=0bar(0bar)
Pullrod Pressure=0bar(0bar)
Pump1=0mA(0mA)
Pump2=0mA(0mA)

	1.Crusher Fwd Swash	50 %
	2.Crusher Rev Swash	50 %
2	3.Fwd Ramp Up Time	1s
2	4.Fwd Ramp Down Time	1s
2	5.Rev Ramp Up Time	1s
2	6.Rev Ramp Down Time	1s
2	7.Max Pressure Point	380 bar
2	8.Over Press Delay Time	10s

Figure 6.8.1 Blockage clearing mode

1—[Main conveyor belt start and stop] button can start or stop the output of the main conveyor belt, which is only valid in this interface. The main conveyor belt must be started before the forward and reverse jog operation of the crusher.

2—[Crusher reverse jog] button, press and hold the button to start the crusher in reverse (counterclockwise), release the button and stop the crusher.

3—[Crusher steering switch] button to determine the steering of the crusher in automatic production mode or manual production mode, which can be switched through this button.

4—[Crusher forward jog] button, press and hold the button to start the crusher forward (clockwise), release the button and stop the crusher.

Note: Before performing the blockage removal operation in this mode, the stones accumulated between the main conveyor belt and the bottom chamber of the crusher should be cleaned first to ensure that there is no stone stuck between the two, so as to avoid causing unpredictable damage to the main conveyor belt after the main conveyor belt is started.

Parameter setting description:

①The forward rotation and reverse rotation of the crusher - the opening of the swash plate - these two parameter values are only used for the jog operation of the crusher. During the process of the crusher blocking and cleaning, it can be adjusted reasonably according to the needs.

②Forward and reverse ramp up time - when the crusher starts jogging, the time required for the proportional valve that controls the opening of the swash plate pump to rise from 0mA to a given mA value, to realize the soft start of the crusher, only suitable for the jog operation of the crusher.

③Forward and reverse ramp down time - when the crusher starts jogging, the time required for the proportional valve that controls the opening of the swash plate pump to drop from a given mA value to 0mA to achieve a soft stop of the crusher, only applicable to the jog operation of the crusher.

④The maximum pressure setting value of the crusher - this parameter is not applicable to the application under the "blocking material removal mode" interface, and is only valid under the "automatic production mode" or "manual production mode" interface. When the actual monitoring pressure of the crushing machine is greater than this set value, and after a certain time delay (parameter 8), the system will pop up an alarm of "high pressure of the crushing machine", and the production process will automatically stop in sequence.

6.9 Folding mode interface

Under the "Main Menu" interface, select the [Folding Mode] button (Figure 6.1.1-6), and it will enter the "Folding Mode" interface, as shown in Figure 6.9.1.

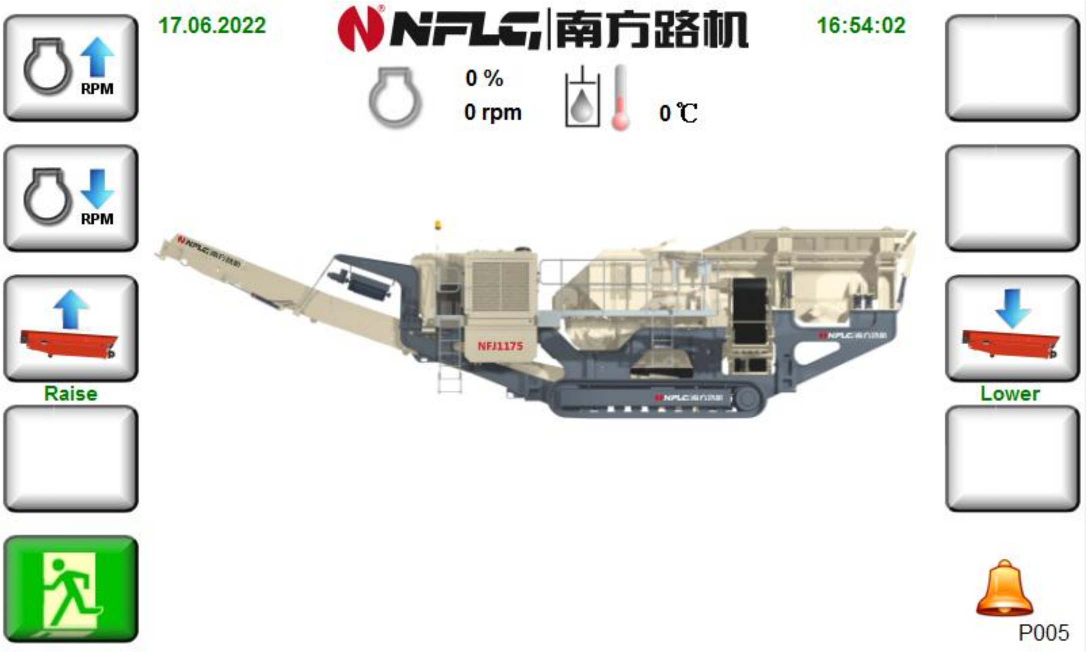
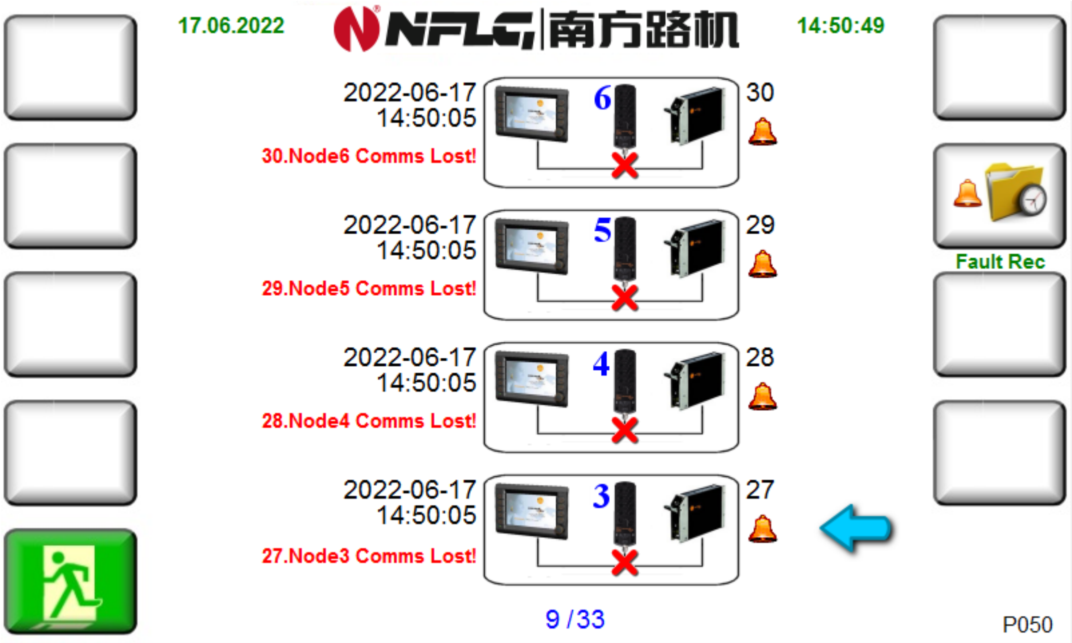

17.06.2022
NFLG|南方路机
16:54:02
RPM
RPM
0 %
0 rpm
0 ℃
NFJ1175
Raise
Lower
P005

Figure 6.9.1 Folding mode interface

This interface can realize the ascending and descending operation of the tail of the main conveyor belt. Before the crawler runs, the tail of the main conveyor belt must be raised to a high position; and before the equipment is produced, the tail of the main conveyor belt must be lowered to a low position.

6.10 Fault information interface

After any fault occurs, no matter which interface it is currently in, it will directly jump to the "Fault Information" interface, as shown in Figure 6.10.1. After exiting this interface, you can enter the "Fault Information" interface again through the [Alarm Information] button under the "Function Information" interface.



17.06.2022 14:50:49

2022-06-17 14:50:05 30.Node6 Comms Lost! 30

2022-06-17 14:50:05 29.Node5 Comms Lost! 29

2022-06-17 14:50:05 28.Node4 Comms Lost! 28

2022-06-17 14:50:05 27.Node3 Comms Lost! 27

9 /33

Figure 6.10.1 Fault information interface

It should be noted:

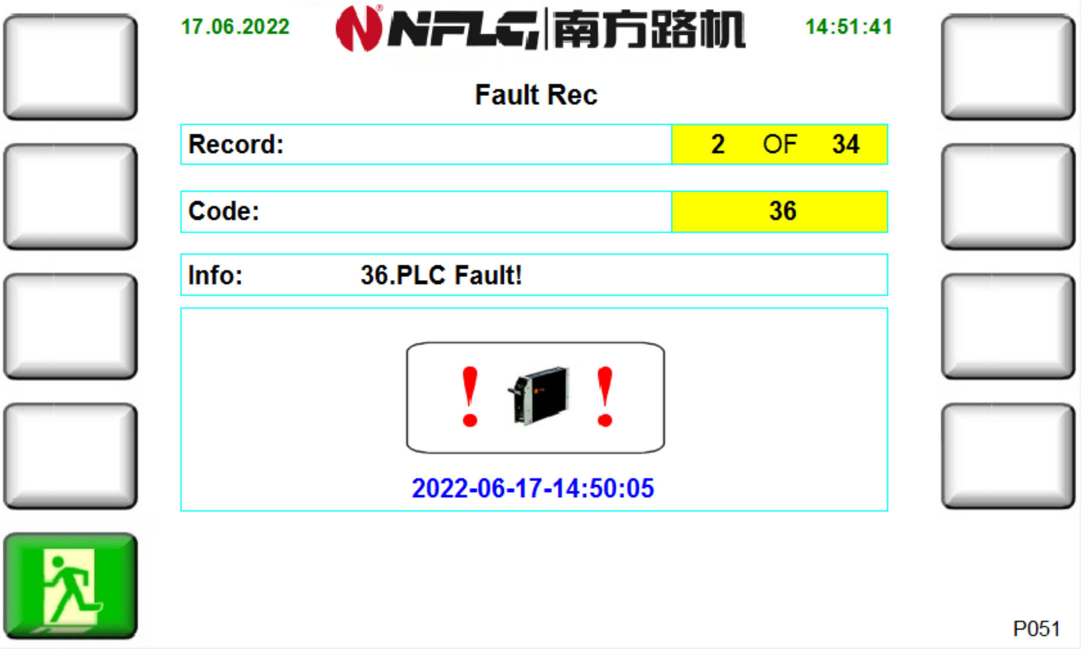
① The entry displayed in the fault information interface is the current fault state of the system. The elimination of each fault entry must satisfy two conditions—the external factors that caused the fault have been eliminated and the fault information needs to be confirmed manually.

There are two states for the elimination of the fault entry - when a fault occurs, if the external factor that caused the fault has been eliminated, the displayed alarm bell icon will change from "🔔" to "🔕", at this time, manual confirmation will be performed(the operation of manual confirmation—>turn the circular knob on the display screen, select the fault, and then press the circular knob), and the fault entry will disappear automatically; when a fault occurs, if a manual confirmation operation is performed first, the fault entry will appear the icon "✓" at the bottom right, indicates that the fault entry has been known to the user. Once the external factors causing the fault are eliminated, the fault entry will disappear automatically.

②[PLC Diagnosis] button will only be displayed when the "PLC fault" alarm occurs in the system. Through this button, you can enter the "PLC Diagnosis" interface to view the fault code or fault information of the current PLC.

6.11 Fault record interface

If there is an unresolved fault currently, you can select the [Fault Record] button in the "Fault Information" interface to enter the "Fault Record" interface. If there is no fault currently, you can select [Alarm Information] in the "Function Information" interface to directly enter the "Fault Record" interface, as shown in Figure 6.11.1. The "fault record" interface displays the latest 200 historical fault information of the system, which will be saved for later inspection or maintenance.



17.06.2022 14:51:41

Fault Rec

Record: 2 OF 34

Code: 36

Info: 36.PLC Fault!

2022-06-17-14:50:05

Figure 6.11.1 Fault record interface

6.12 Engine information interface

In the "Function Information" interface, select the [Engine Information] button to enter the "Engine Information" interface, as shown in Figure 6.12.1. The engine ECU transmits the basic data to the on-board controller through J1939 communication. When the on-board controller communicates with the display screen normally, the data displayed on this interface will be synchronized with the data monitored by the engine ECU.

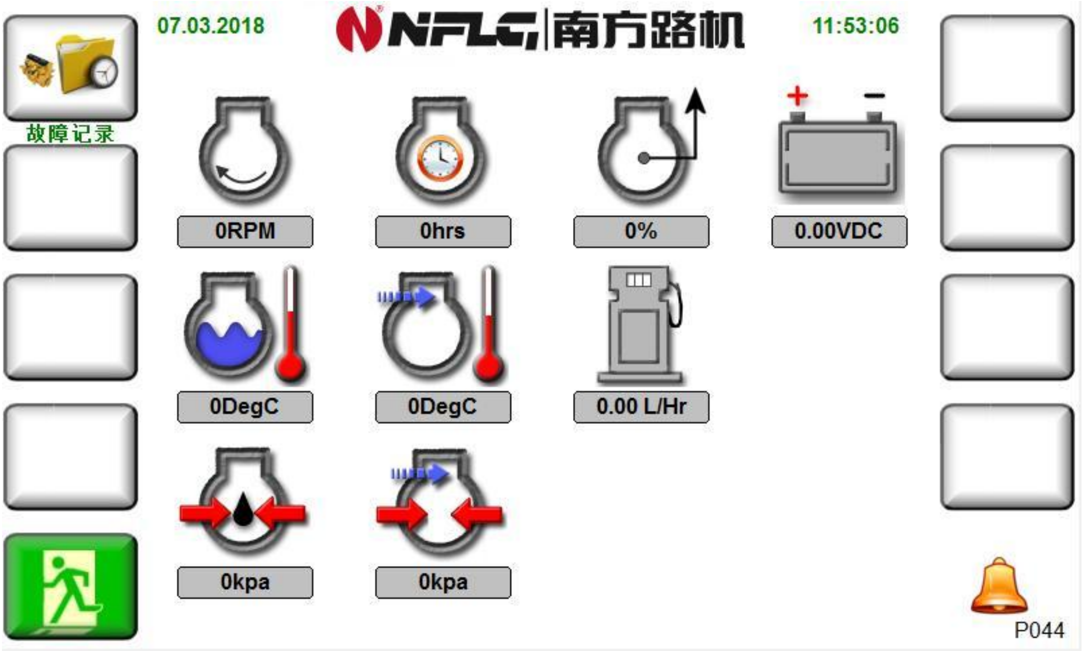
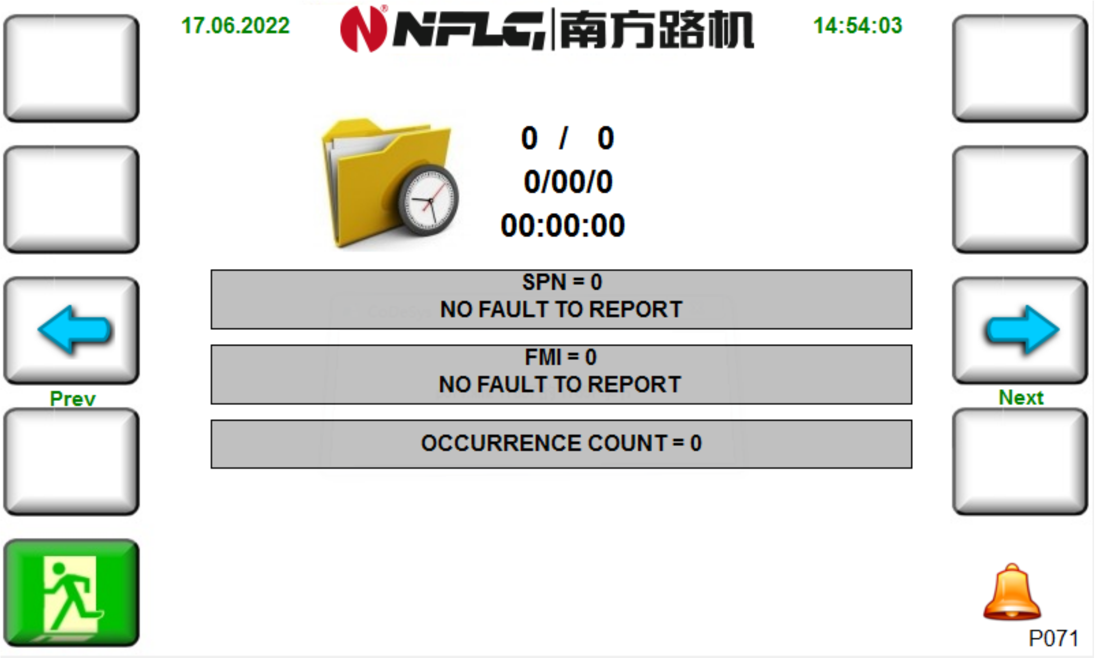


Figure 6.12.1 Engine information interface

The meanings of the icons in the interface are from left to right, and from top to bottom are engine speed RPM, engine operating hours hrs, engine load %, ECU voltage VDC, engine coolant temperature DegC, intake manifold temperature DegC, engine instantaneous fuel consumption L/Hr, oil pressure Kpa, intake manifold pressure Kpa.

6.13 Engine failure history interface

In the "Engine Information" interface, select the [Fault Record] button to enter the "Engine Fault Record" interface, as shown in Figure 6.13.1. If an "engine failure" alarm occurs in the system, you should enter this interface to check the specific failure information of the engine. You can search the engine maintenance manual through the SPN code and FMI code to obtain more detailed failure descriptions. If necessary, please contact the engine manufacturer for handling.



0 / 0
0/00/0
00:00:00

SPN = 0
NO FAULT TO REPORT

FMI = 0
NO FAULT TO REPORT

OCCURRENCE COUNT = 0

Figure 6.13.1 Engine failure history interface

6.14 Application selection interface

In the "Automatic Production Mode" or "Manual Production Mode" interface, select the [Application Selection] button to enter the "Application Selection" interface, as shown in 6.14.1.

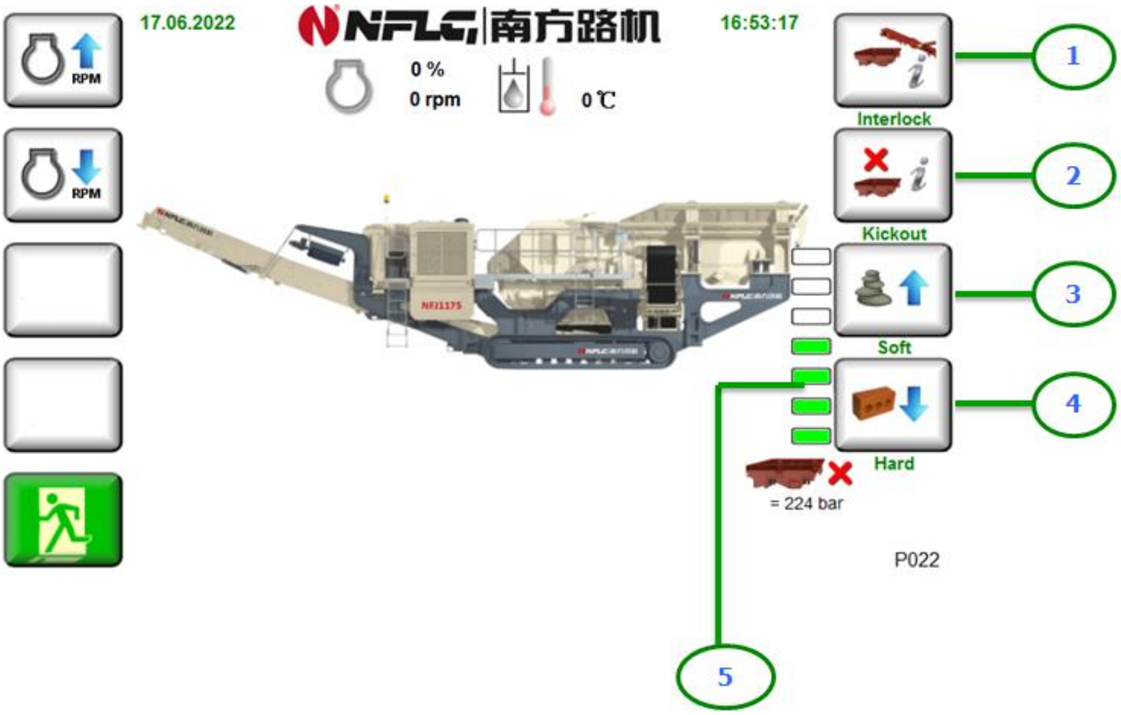


Figure 6.14.1 Application selection interface

1—Enter the interface of "upper and lower chain control parameters".

2—Enter the interface of "Pause Parameters of Feeder", see Figure 6.3.5.

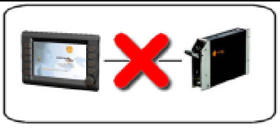
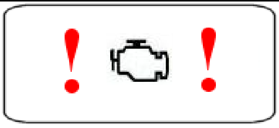
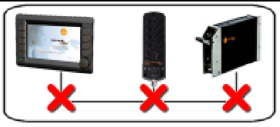
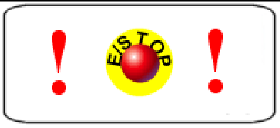
3—[Soft material] selection button   4—[Hard material] selection button

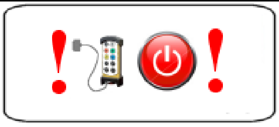
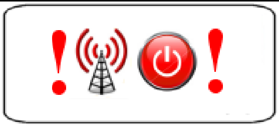
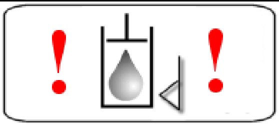
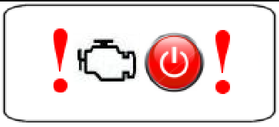
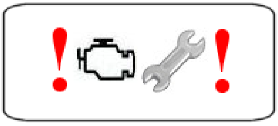
5—Status display of the selected material application type

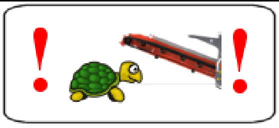
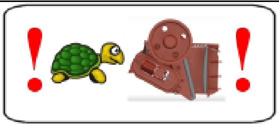
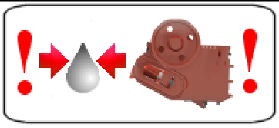
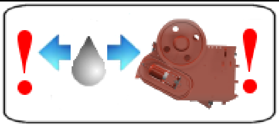
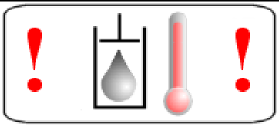
For different stone characteristics and application occasions, the system has seven application types to choose from. For the corresponding control parameter setting values, please refer to the above-mentioned "Feeder Pause Parameters" interface parameter setting description. It can be seen from the status display of the application type: the softer the material, the lower the pressure of the crushing host of the suspended feeder, on the contrary, the harder the material, the greater the pressure of the crushing host of the suspended feeder. The operator can choose according to the actual situation of the site to achieve the best performance of the equipment.

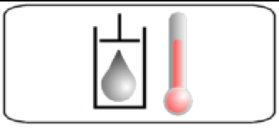
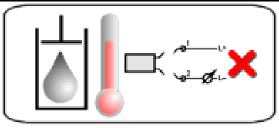
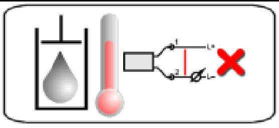
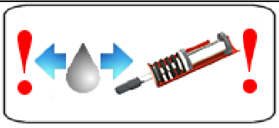
7 Fault tracing

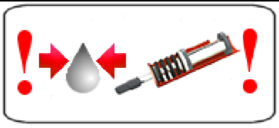
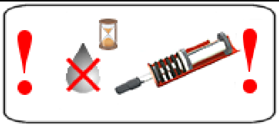
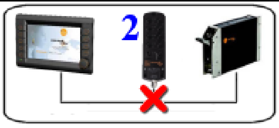
The following table lists some common fault items of equipment, and gives possible causes of faults and reference solutions. For further assistance, please contact our company.

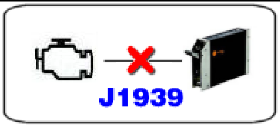
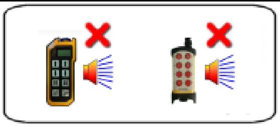
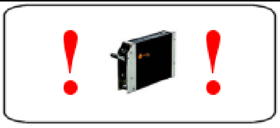
Error code	Possible reason	Solution	Fault Icon & Description
1	♦ The communication line between the PLC and the display screen is incorrectly wired or not connected.	✓ Check the wiring to make sure that CANH and CANL are properly connected.	 The connection between the display screen and the PLC is interrupted!
	♦ The CANBUS network terminating resistor is not connected or installed incorrectly.	✓ Check the resistance value on the CAN network in the power-off state, it should be 60 ohms.	
	♦ The PLC is faulty or the program is not running.	✓ Check the PLC status light, normally it flashes green at 2Hz.	
2	♦ The engine ECU has detected some kind of malfunction in itself.	✓ Check the specific SPN code and FMI code of the engine failure to obtain more detailed failure information, and contact the manufacturer in time.	 Engine failure!
3	♦ The CANBUS bus plug on the side of the cabinet is not connected or loose.	✓ Check the CAN bus plug on the side of the cabinet to make sure it is connected properly.	 CANBUS bus error!
	♦ The CANBUS network terminating resistor is not connected or installed incorrectly.	✓ Check the resistance value on the CAN network in the power-off state, it should be 60 ohms.	
	♦ A node on the network has an incorrect baud rate setting.	✓ Check the baud rate settings of each node to ensure that the baud rates are consistent.	
5	♦ The emergency stop button of the equipment is pressed.	✓ Check the condition of all emergency stop buttons on the equipment, and turn right to reset.	 Equipment emergency stop failure!
	♦ The emergency stop button contacts are damaged or have poor contact.	✓ In the power-off state, check the on-off status of each emergency stop button contact separately, and replace the emergency stop button contact of the same type if necessary.	

	♦ Damaged safety relay or poor contact.	✓ Replace the safety relay of the same type.	
6	♦ The wired remote control plug or emergency plug is not connected.	✓ Check the wired remote control plug or emergency plug on the side of the cabinet to make sure it is properly connected.	 Wired remote control emergency stop failure!
	♦ The emergency stop button of the wired remote control is damaged.	✓ Replace the emergency stop button of the same model.	
	♦ The emergency stop button of the wired remote control is pressed.	✓ Check the emergency stop button on the remote control and rotate to reset.	
7	♦ The emergency stop button of the wireless remote control is damaged.	✓ Please send the wireless remote control back to the manufacturer for processing.	 Wireless remote control emergency stop failure!
	♦ The emergency stop button is pressed while the wireless remote control is active.	✓ Check the emergency stop button on the remote control, pull it out to reset.	
	♦ The wireless remote automatically powers off when activated.	✓ When the wireless remote control is not in use, please turn off the wireless remote control function in time.	
8	♦ The hydraulic oil level is low.	✓ Add suitable hydraulic oil in time.	 The hydraulic oil level is low!
	♦ Damaged hydraulic oil level sensor.	✓ Replace the lubricating oil level sensor of the same type.	
	♦ Hydraulic oil level sensor wiring is loose or not connected.	✓ Check the wiring to make sure the hydraulic oil sensor wiring is correct and secure.	
10	♦ Engine speed is below 300rpm.	✓ It may be caused by too much external load. Please check carefully and find out the reason according to the actual situation on site.	 Engine emergency stop!
11	♦ The engine has been running for 250 hours without maintenance.	✓ Please strictly abide by the requirements of the Caterpillar engine maintenance manual, and carry out comprehensive maintenance and	 Engine maintenance expired!

		maintenance of the engine on a regular basis.	
12	♦The speed wheel sensor wiring is loose or not connected.	✓Check the wiring to make sure the speed wheel sensor wiring is correct and secure.	 Main conveyor speed is low!
	♦Damaged speed wheel sensor.	✓Replace the speed sensor of the same specification.	
	♦The main belt conveys too much material or the main belt is stuck.	✓Remove material from the belt; check the main belt mechanics for the cause of the jam.	
	♦The speed pulley is not in proper contact with the belt.	✓Adjust the installation position of the speed pulley to ensure that the speed pulley is in correct contact with the belt.	
13	♦The host speed proximity switch is damaged.	✓Replace the speed proximity switch of the crusher of the same specification.	 The speed of the crusher is low!
	♦The crushing cavity is blocked or there are unbreakable materials.	✓Avoid over-sized materials or unbreakable materials from entering the crusher.	
	♦The speed proximity switch is not installed correctly and no signal is detected.	✓Adjust the installation position of the proximity switch, the detection distance should be less than 6mm.	
14	♦The crusher chamber is blocked or there are unbreakable materials.	✓Avoid over-sized materials or unbreakable materials from entering the crusher.	 The crushing host has high pressure!
	♦If there is no material in the chamber, there may be the interference with the relevant mechanical structure, or there may be a problem with the hydraulic system.	✓Check the relevant mechanical or hydraulic parts to find out the cause.	
15	♦The oil replenishment pump is damaged and cannot replenish oil to the hydraulic circuit of the crusher, and the pressure is lower than 15Bar	✓Replace the charge pump as soon as possible.	 Crusher low pressure!
18	♦Hydraulic oil temperature is above critical setpoint.	✓Stop and wait until the hydraulic oil temperature drops. If this fault occurs	

		frequently, it must be checked mechanically or hydraulically. ✓Check whether the hydraulic oil cooling fan is working normally and whether the filter element is blocked.	High hydraulic oil temperature shutdown!
19	♦The hydraulic oil temperature is higher than the alarm set value.	✓This alarm only prompts the operator and does not do any chain control. If it is found that the temperature rises too fast, you should pay attention and stop the machine to check the mechanical or hydraulic parts if necessary. ✓Check whether the hydraulic oil cooling fan is working normally and whether the filter element is blocked.	 Hydraulic oil temperature high alarm!
20	♦The wiring of the hydraulic oil temperature sensor is loose or not connected, and the analog signal is lower than 3.5mA.	✓Check the sensor wiring to make sure it is correct and secure.	 Hydraulic oil temperature sensor open circuit!
	♦The hydraulic oil temperature sensor is damaged, and the analog signal is lower than 3.5mA.	✓Replace the temperature sensor with the same specification.	
21	♦The hydraulic oil temperature sensor is damaged, and the analog signal is higher than 21mA.	✓Replace the temperature sensor with the same specification.	 Hydraulic oil temperature sensor short circuit!
	♦The hydraulic oil temperature sensor circuit is short-circuited, and the analog signal value is greater than 21mA.	✓Check the wiring and replace the entire casting line if necessary.	
22	♦There is a leak in the tie rod cylinder or in the connected oil line.	✓Check the relevant hydraulic parts to make sure there are no oil leaks.	 Low lever pressure! (The troubleshooting of this fault also applies to fault code 25)
	♦The plug of the filling solenoid valve is energized, and the valve core cannot be opened normally.	✓Check the solenoid valve spool or the entire tie rod hydraulic valve block.	
	♦The throttle valve of the tie rod valve block is not closed tightly.	✓Close the throttle valve on the tie rod valve block as required.	

	♦The fill solenoid valve is not energized.	✓Check the wiring to make sure that the wiring is correct and secure, and that the cables are not damaged.	
24	♦The filling solenoid valve is de-energized and the spool cannot close properly.	✓ Check the solenoid valve spool or the entire tie rod hydraulic valve block.	 Tie rod pressure is high!
	♦The fill solenoid valve remains energized continuously.	✓ Make sure that the upper limit of the tie rod charging pressure is set correctly.	
	♦The hydraulic flow is too large, and too much oil enters the tie rod instantly.	✓Please check and adjust the hydraulic related parts.	
25	♦The throttle valve of the tie rod valve block is not closed tightly.	✓Close the throttle valve on the tie rod valve block as required.	 Tie rod filling time out! (The troubleshooting of this fault also applies to fault code 22)
	♦The relief valve pressure setting of the tie rod valve block is incorrect.	✓Set the relief valve pressure of the tie rod valve block as required.	
	♦There is a leak in the tie rod cylinder or in the connected oil line.	✓Check the relevant hydraulic parts to make sure there are no oil leaks.	
	♦The hydraulic flow is too small, and too little oil enters the tie rod within the specified time.	✓Please check and adjust the hydraulic related parts.	
26	♦The bus plug of the node module is not connected or is loose.	✓Check the corresponding bus plug on the T-adapter to make sure the connection is normal and secure.	 Node module 2 connection interrupted! (Refer to this section for troubleshooting other node module connection interruptions)
	♦Node module is not powered.	✓Check the wiring or power supply, the node module requires DC24V power supply.	
	♦The address of the node module is not set correctly.	✓ The node module address must be consistent with the electrical schematic.	
	♦A channel cable failure or input/output failure on the node module.	✓Remove the plugs in sequence from channels 1-8 of the node module, attempting to reset the fault each time one is removed.	
30	♦The engine ECU is not powered or the power supply is	✓Check the power wiring of the ECU to ensure that the	

	abnormal.	ECU plug is securely connected.	 J1939 connection interrupted!
	♦ The communication wiring between the engine and the PLC is wrong or not connected.	✓ Check the communication wiring to make sure that CAN2H and CAN2L are connected correctly.	
	♦ The J1939 network terminating resistor is not connected or installed incorrectly.	✓ When the power is off, check the resistance value on the network, it should be 60 ohms.	
35	♦ The wired remote control and wireless remote control functions are selected and activated at the same time.	✓ Wired remote and wireless remote cannot be used at the same time, you must deselect both remotes and re-enable a single remote.	 Do not use wired and wireless remote control at the same time!
36	♦ VBBR power failure.	✓ Check system voltage.	 PLC failure!
	♦ I/O failure.	✓ Determine which channel is faulty based on the information from the PLC diagnostic interface, check and find out the cause of the fault.	

User feedback form

Device name:

Machine number: Electrical drawing number:

NFLG appreciates your valuable comments. Please answer the following questions, choosing one option for each question.
Your opinion will improve our equipment!

How would you rate the quality of this device?

	Superb	Good	Ordinary	Bad
General impression	☐	☐	☐	☐
Structural design	☐	☐	☐	☐
Stability	☐	☐	☐	☐
Operability	☐	☐	☐	☐
Control method	☐	☐	☐	☐
Human-machine interface	☐	☐	☐	☐

Suggestion:

How would you rate the quality of the information booklet?

	Superb	Good	Ordinary	Bad
General impression	☐	☐	☐	☐
Layout design	☐	☐	☐	☐
Schematic callout	☐	☐	☐	☐
Readability	☐	☐	☐	☐
Easy search	☐	☐	☐	☐
Content structure	☐	☐	☐	☐

Suggestion:____

Other suggestion:____________________________________

Please return the User Feedback Form to the following address:

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Province No. 700 Sports Street 362000
Fujian South Highway Machinery Co.Ltd
Electrical division

Sincere thanks!

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